Transportation Research Part E 165 (2022) 102839

Contents lists available at ScienceDirect

Transportation Research Part E

journal homepage: www.elsevier.com/locate/tre

Anti-Counterfeiting in a retail Platform: A Game-Theoretic approach

Yu Zhou [a,b], Xiang Gao [a], Suyuan Luo [c,*], Yu Xiong [d], Niangyue Ye [e]

[a] School of Economics and Business Administration, Chongqing University, Chongqing, China
[b] Chongqing Key Laboratory of Logistics at Chongqing University, Chongqing, China
[c] Department of Transportation Economics and Logistics Management, College of Economics, Shenzhen University, Shenzhen, China
[d] Surrey Business School, University of Surrey, Surrey, UK
[e] Shenzhen Technology University, Shenzhen, China

ARTICLE INFO

Keywords:
Anti-counterfeiting
Dual channel
Retail management
Platform economics
Game theory

ABSTRACT

The retail platform has developed rapidly, but the problem of fake products has also become increasingly severe. This paper investigates the impact of anti-counterfeiting in a retail platform and the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology by using a game-theoretic model. We consider that the product can be sold directly by the manufacturer, or indirectly through a reseller on the platform. The reseller might also sell fake products, but the platform and the manufacturer can use anti-counterfeiting technology to fight against the fakes. Our analysis shows that the payoff of anti-counterfeiting in the retail platform is not always positive. Specifically, when the production valuation is low, the anti-counterfeiting payoff for the platform (the manufacturer) is negative if the proportion of fakes is sufficiently low (high). We also find that anti-counterfeiting may harm consumer surplus and social welfare. In addition, if the investment cost of anti-counterfeiting is high, at most one firm, either the platform or the manufacturer, has the incentive to invest in anti-counterfeiting contingent on the relative valuation on the platform's services. Finally, with the investment in anti-counterfeiting, the platform should provide better services than before for surviving in the market.

1. Introduction

With the continuous improvement of the services provided by the retail platform, more and more consumers like to shop on the platform (Cao et al., 2019; Choi and He, 2019; Wang et al., 2019; Song et al., 2020). At the same time, in order to broaden distribution channels, many brand manufacturers sell their products to the resellers on retail platforms in addition to their direct sales channels (Hagiu and Wright, 2015; Li et al., 2019; Yan et al., 2020). For example, many apparel brands, such as Adidas, Nike, and Li Ning, not only sell directly through their own channels, but also wholesale their products to the resellers on retail platforms, such as JD.com, Taobao, and Pinduoduo. With the pursuit of consumers and the implementation of the manufacturers' dual channel strategy, the retail platform has developed rapidly (Choi et al., 2020a; Xu and Choi, 2021). In 2020, the value of the global B2C e-commerce market was US$3.67 trillion.[1] Under such bright prospects, the retail platform is also facing a problem, that is, resellers may counterfeit products

* Corresponding author.
E-mail address: suyuanluo@126.com (S. Luo).

[1] https://www.grandviewresearch.com/industry-analysis/b2c-e-commerce-market. [Accessed on 6 Nov 2021].

https://doi.org/10.1016/j.tre.2022.102839
Received 30 November 2021; Received in revised form 23 June 2022; Accepted 21 July 2022
Available online 5 August 2022
1366-5545/© 2022 Elsevier Ltd. All rights reserved.

after purchasing authentic products from manufacturers, and sell both fake products and authentic products to consumers on the platform. It was reported that most fast moving consumer goods on the platform are sold by resellers, some of which mixed fake products into authentic products.[2]

At present, the problem of fake products on a retail platform is very prominent. In 2016, The Counterfeit Report (TCR), an advocacy organization that cooperates with real producers to examine fakes, has identified about 58,000 fake products on Amazon, and TCR stated that the total number of counterfeits on Amazon might far exceed 58,000 because this only involves the brands it cooperates with.[3] As fake goods are inferior to the authentic ones, and resellers mixing authentic and fake goods is a kind of deception, which will harm the interests of consumers and society. As Steve Francis, the Assistant Director for HIS's Global Trade Investigations Division, said that fake goods are growing at an alarming rate and the performance of these fake goods may fail catastrophically.[4] In this context, many retail platforms and manufacturers have begun to take anti-counterfeiting action. However, traditional anti-counterfeiting measures are difficult to accurately identify the authenticity of the products, and are unable to completely eliminate counterfeits.

Recently, the development of blockchain technology has made traceability of the supply chain possible so that counterfeit products can be effectively identified and eliminated (Choi and Luo, 2019; Shen et al., 2020; Li et al., 2022; Niu et al., 2022). What makes blockchain technology different from other anti-counterfeiting technologies is that it allows consumers to have complete confidence in the authenticity of products due to its immutability and traceability (Babich and Hilary, 2020; Pun et al., 2021; Cao and Shen, 2022; Shen et al., 2022). Therefore, many platforms are using blockchain technology to eliminate fakes in order to completely eliminate consumers' concerns about fakes and increase consumers' willingness to buy products on their platforms. For example, eBay, Tmall, Everledger, and Lazada have already developed blockchain to keep permanent record of reliable data to help consumers verify the authenticity of products (Choi, 2019; Niu et al., 2021). On Taobao, fashion labels are required to upload their designs to a blockchain technology system that will examine similar products automatically, and the identified fakes will be deleted from the platform.[5] To eliminate the cannibalization effect of fake products, many manufacturers are also using blockchain technology to crack down on fakes. Luxury giant LV and diamond company Chow Tai Fook help consumers identify the authenticity of products by disclosing product information through a blockchain system, and Chronicled implements blockchain to verify authenticity in the drug supply chain (Pun et al., 2021). At the same time, manufacturers also use blockchain technology to combat counterfeits with the help of retail platforms. Walmart help food suppliers Dole Food and Driscoll's to implement blockchain to identifies product quality; Nestlé is developing blockchain technology with the help of Amazon that timely presents information about the specific production and process of the beans of new coffee brands, helping consumers distinguish between genuine and fake products (Shen et al., 2022).

Anti-counterfeiting has an impact on two aspects. First, from the perspective of competition in the retail market, anti-counterfeiting makes the reseller no longer sell fake products; that is, all products that the reseller sells are authentic products. Although the reseller can set a higher price, this makes its products less competitive. Second, from the perspective of incentive for reselling, anti-counterfeiting makes it impossible to carry out low-cost counterfeit products, which increases the marginal cost of resale, reducing the incentive for resale. Thus, for the retail platform, on the one hand, anti-counterfeiting can eliminate counterfeit products and increase the retail price of products sold on the platform so that the platform may obtain more commissions. However, on the other hand, anti-counterfeiting may also reduce the competitiveness and incentive for reselling, thus reducing the sales of the reseller so that the platform may also receive fewer commissions. For the manufacturer, intuitively speaking, anti-counterfeiting should bring positive revenue. However, since the revenue of the manufacturer comes from the direct channel and resell channel, the impacts of anti-counterfeiting on the manufacturer are more complicated. Anti-counterfeiting can certainly eliminate the cannibalization of counterfeit products on the direct channel, but after eliminating the counterfeit products, the reseller's incentive for reselling may decrease, which could be a potential negative for manufacturers. In such a context, we first aim to answer the following research question:

Will the platform and the manufacturer always obtain positive revenues from anti-counterfeiting?

With the cost of anti-counterfeiting technology, the platform and manufacturer may not be able to afford this cost. Thus, it is not clear who will have the incentive to invest in anti-counterfeiting technology. Therefore, from the perspective of the investment cost of the anti-counterfeiting technology, we discuss the following research question:

Who will have the incentive to invest in anti-counterfeiting technology, the platform, or the manufacturer?

In addition, the retail platform usually provides services that are different from those of the manufacturer. This differentiated service will affect consumers' purchasing choices, which may change market competition and change the impact of anti-counterfeiting, but this change is not intuitive. Taking into account the increasing pressure of anti-counterfeiting, the platform may have to use anti-counterfeiting technology to eliminate counterfeit goods. Therefore, we further discuss the following research question:

After anti-counterfeiting, how should the retail platform adjust the services it provides?

We establish a dual-channel competition model that the product can be sold directly by the manufacturer, or indirectly through a reseller on the platform. We also consider the consumer's channel preference and channel service differences. Based on the model, we solve the sub-game equilibria in the cases with and without anti-counterfeiting. By comparing these equilibria, we obtain the impact of anti-counterfeiting and the incentives for platform and manufacturer to invest in anti-counterfeiting technology. To our knowledge, our work is the first to study anti-counterfeiting in a retail platform under dual-channel competition.

[2] https://www.kepuchina.cn/qykj/qytt/201611/t20161107_43811.shtml. [Accessed on 11 Apr 2022].
[3] https://www.orderhive.com/how-to-fight-amazon-counterfeit-listings-and-what-is-project-zero. [Accessed on 6 Nov 2021].
[4] https://www.ice.gov/features/dangers-counterfeit-items?msclkid=27593872b49c11ecaca5ed6e65e9c44b. [Accessed on 13 Apr 2022].
[5] https://jingdaily.com/luxury-brands-china-intellectual-property/. [Accessed on 6 Nov 2021].

Our analysis mainly shows the following results. First, the platform and the manufacturer do not always obtain a positive revenue from anti-counterfeiting. For the platform, when the proportion of the fake products and consumers' valuation on the products are both low, the anti-counterfeiting revenue of the platform is negative. This is because if the proportion of the fake product and consumers' valuation on the product are both low, anti-counterfeiting will not increase consumers' valuation on platform products much. Although the retail price of the reseller will increase, the market demand for the reseller will drop a lot, resulting in a drop in the commission that the platform can charge. For the manufacturer, intuitively, the anti-counterfeiting revenue can be positive. However, we find that when the proportion of the fake product on the platform is high, the anti-counterfeiting revenue of the manufacturer is negative. This is because when the proportion of the fake product on the platform is high, the manufacturer can set a high wholesale price, but anti-counterfeiting will make the wholesale price decrease sharply, leading to the decline of the revenue of the manufacturer. Moreover, we find that anti-counterfeiting may harm consumer surplus and social welfare.

Second, the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology are not consistent. If the investment cost of anti-counterfeiting is relatively high and the consumers' valuation on the platform's services is high, only the platform has the incentive to invest in anti-counterfeiting technology. This is because, many consumers will buy the product sold by the reseller if the consumers' valuation on the platform's services is large, so the platform has more incentive to anti-counterfeiting to make those consumers to pay a higher retail price. However, if the investment cost of anti-counterfeiting is relatively high and the consumers' valuation on the platform's services is low, only the manufacturer has the incentive to invest in anti-counterfeiting technology. This is because, in this case, the number of the consumers who buy the product from the reseller in the platform is small. The raised retail price in the platform in the presence of anti-counterfeiting will further reduce the demand in the resell channel. Thus, the platform is reluctant to invest in anti-counterfeiting.

Third, when anti-counterfeiting technology is used to drive fake products out of the market, the platform should provide better platform services. With investment in anti-counterfeiting, consumers are willing to pay a high price for the product sold by the reseller. If the platform could provide better platform services, then the number of consumers willing to pay a high price for the product sold by the reseller will rise.

The remainder of this paper is structured as follows. Section 2 reviews the related literature, followed by the description of the model in Section 3. In Section 4, we present the decision of the manufacturer and the reseller in the case without anti-counterfeiting and in the case with anti-counterfeiting, then analyze the impacts of anti-counterfeiting and the incentives for platforms and manufacturers to invest in anti-counterfeiting technology, as well as the impact of anti-counterfeiting on consumer surplus and social welfare. For robustness checking of the core results, we consider the extended model in Section 5. In the final section, the study is concluded with suggestions for further research. All mathematical proofs are collected in the Appendix.

2. Literature

Two streams of literature are related to our work: (1) retail channel, especially dual channel and retail platform, and (2) counterfeiting and anti-counterfeiting. To our best knowledge, our work is the first one focusing on the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology under dual-channel competition

2.1. Retail channel

A wealth of results has been obtained in the research on the retail channel, and the research related to this paper mainly involves dual-channel and retail platforms. We will review these two streams of literature.

In order to attract more consumers, it is common for firms to adopt a dual-channel distribution strategy. The competition and cooperation relationship under the dual-channel structure, the purchase behavior of consumers, and the pricing strategy of different channels will significantly affect the profit of the firm. In terms of competition and cooperation under dual channels, Xiao and Choi (2009) analyzed the quantity competition between the distribution channels of two manufacturers, where each manufacturer can sell a product through either an integrated channel or a retailing channel. Liu et al. (2016) investigated the price competition under a dual channel for the WEEE recycling market under the impact of government subsidy. Moreover, Liu et al. (2017) demonstrated the impact of retailer alliances on the profit of the manufacturer under the dual-channel structure. Choi et al. (2019a) built an online-to-offline fashion franchising model and analyzed when channel conflicts are avoided. In terms of consumer purchase behavior, relevant research considers the impact of different consumer behaviors on the dual channel, such as consumer free-riding (Liu et al., 2020a), consumer network acceptance (Liu et al., 2020c), and online consumer reviews (Yang et al., 2021). Regarding pricing strategy, relevant literature reported the impact of different factors on pricing strategies under the dual channel, such as supply shortage (Xiao and Shi, 2016), trade credit policy (Qin et al., 2020a), and information disclosure (Zhou et al., 2020). In this paper, we study both the impact of competition between direct channel and resell channel and the impact of consumers' purchase choices on the pricing strategy under the dual channel. Different from the above-mentioned literature, we also take into account the differences in channel services. This paper shows that the differences in channel services will have an important impact on consumers' purchasing choices.

With the popularity and development of the Internet, there are more and more retail platforms (Choi et al., 2019b; Luo et al., 2022). As an online sales mode, the retail platform is in sharp contrast with a traditional physical store. Many scholars have conducted comparative studies, such as Arya and Mittendorf (2018), Yang and Tang (2019), He et al. (2021). Based on the retail platform, a variety of sales modes and strategies have been derived, such as the marketplace mode and the reseller mode. Numerous studies have been conducted, such as Zhang et al. (2019), Liu et al. (2020b), Wei et al. (2020). There are also studies focusing on hybrid sales modes on the retail platform. For instance, Tian et al. (2018) studied the hybrid mode of the marketplace and reseller and compared the pure

reseller mode with the pure marketplace mode. Qin et al. (2020b) investigated the logistics service sharing strategy of a hybrid online platform that acts both as a retailer and a marketplace service provider. There are also some work about the retail platform services. Zennyo (2020) examined the strategic contracting between an e-commerce service platforms and suppliers. Song et al. (2021) explored how consumer awareness affects the third parties' selling strategies and retailers' platform openness, in which the retailer's platform has valuation advantage due to better consumer service. Siqin et al. (2022) investigated the optimal channel selection and service contracting in the e-platform. In addition, research has been conducted on retail platform from the perspective of information asymmetry, such as Kwark et al. (2017), Dong et al. (2018), Zhang and Zhang (2020). While the above literature mostly focuses on different sales modes based on the retail platform, we consider the competition between the resell channel based on the retail platform and the direct channel of the manufacturer, as well as the channel preferences of consumers.

2.2. Counterfeiting and anti-counterfeiting

It is often difficult for consumers to judge the authenticity of products, and counterfeit products may have a price advantage. Therefore, the existence of counterfeit will compete with authentic products to a certain degree. Many studies have been conducted on this competitive relationship and its impact, such as Qian et al. (2015), Gao et al. (2017a), Pun and DeYong (2017). Zhang and Zhang (2015) examined the optimal supply chain structure when counterfeit products exist. The results show that when the distribution channel (i.e., general channel) is penetrated by counterfeit products, brand companies may need to reorganize their product distribution structure and rely on reliable channels to ensure 100% authenticity. Gao et al. (2017b) studied the competition between imitations and incumbent products with a two-period dynamic non-cooperative game model. The results show that the entry of potential imitators brings (hidden) pressure on the incumbents, forcing them to lower their prices, and thus possibly beneficial to consumer welfare. Different from the above literature, we assume that the reseller selling counterfeits needs to purchase genuine products from the manufacturer. Therefore, the reseller and the manufacturer not only have a competitive relationship, but also have a cooperative relationship within the supply chain, which complicates the game relationship between the manufacturer and the reseller and the impact of anti-counterfeiting.

As the competition between counterfeits and authentic products may cause damage to firms that manufacture and sell authentic products, there are current documents that focus on how to combat counterfeit or prevent counterfeit products, such as Cho et al. (2015), Gao (2018), Yao and Zhu (2020). Cui (2019) studied the effect of the original equipment manufacturer's investment strategy in quality improvement on preventing the contract manufacturer from imitation and encroachment. The results show that the structure of quality improvement has a strong impact on preventing encroachment. Yi et al. (2020) focused on a supply chain composed of a manufacturer and a retailer and studied how the supply chain can effectively combat counterfeit products. The results show that the manufacturer is more inclined to lower the wholesale price to induce the retailer to combat counterfeit products rather than fight against itself. Ghamat et al. (2021) studied how a manufacturer can prevent counterfeits through vertical integration and intellectual property agreements when a supplier and third party may produce counterfeit products. They concluded that the manufacturer's decision to sign an intellectual property agreement could benefit the supplier. With the new technology of blockchain, anti-counterfeiting is more efficient and convenient (Choi et al., 2019c; Choi and Ouyang, 2021). Literature has reported the studies on the use of blockchain technology to combat counterfeiting, such as Choi (2019), Choi et al. (2020b), Niu et al. (2021). Pun et al. (2021) studied how blockchain technology can be used by firms and governments to fight against counterfeiting from the perspective that the adoption of blockchain technology transforms a deceptive counterfeit setting into a non-deceptive counterfeiting setting. The results reveal that the government's subsidy for blockchain technology will be more beneficial to consumers and society than the government's enforcement policies. The above-mentioned literature studied different anti-counterfeiting strategies, but most of the literature only considered the manufacturer's anti-counterfeiting decisions. In contrast, with the reality that retail platforms are also facing increasing pressure to crack down on counterfeiting, we study the anti-counterfeiting incentives of the retail platform and the manufacturer under dual channel competition.

Currently, there are only a few studies on anti-counterfeiting on retail platform. Choi (2019) examined the values of blockchain technology supported platforms for diamond anti-counterfeiting in luxury supply chains. Shen et al. (2022) investigated how a permissioned blockchain technology platform combats copycats in the supply chain and how it benefits brand name companies. However, the above literatures consider the platform acts as a retailer, while we consider the platform acts as a marketplace. Sun et al. (2020) explored the interactions between an authentic brand seller and a counterfeiter of the brand on an online marketplace. But this paper assumes that both the manufacturer and the counterfeiter are selling on the platform, and there is only competition between the counterfeiter and the manufacturer. Different from this paper, we consider that the product can be sold directly by a manufacturer, or indirectly through a reseller on the platform, but the reseller might also sell fake products. So, in our work, only the reseller selling on the platform, and the reseller not only compete with the manufacturer but also has a supply chain relationship with the manufacturer.

2.3. Our contributions

By bridging these two streams, our work focuses on the impact of anti-counterfeiting in a retail platform and the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology under dual channel competition. Our study makes the following contributions to the literature.

First, the existing anti-counterfeiting literature mostly only considers anti-counterfeiting decisions from the perspective of manufacturers, while we focus on the anti-counterfeiting incentives of the retail platform and the manufacturer. Our study reveals that if the investment cost of anti-counterfeiting is low, both the platform and the manufacturer are willing to fight against counterfeiting;

however, if the investment cost of anti-counterfeiting is relatively high, only the platform (the manufacturer) is willing to fight against counterfeiting when the consumers' valuation on the platform's services is large (small).

Second, the existing anti-counterfeiting literature mostly only considers the competition between manufacturers and imitators. In our model, the reseller sells counterfeit goods in the retail market, thus forming a competitive relationship with the manufacturer; but the reseller also purchases genuine products from the manufacturer, and thus also has a supply chain partnership with the manufacturer. We find that this complex game relationship between the reseller and the manufacturer may make the manufacturer suffer from anti-counterfeiting.

Third, after analyzing the impact of the consumers' valuation on the platform's services and the proportion of the fake product on the revenues of the manufacturer and the reseller and the platform, we find that the manufacturer, the reseller, and the platform may all benefit from the increase of the consumers' valuation on the platform's services, while the impact of the proportion of the fake product on revenues is non-monotonic.

3. Model

In this study, a game-theoretic model is developed to analyze the implications of anti-counterfeiting in a retail platform. A manufacturer (referred to as she) produces a product at a unit production cost of$c$. The product can be sold directly by the manufacturer, or indirectly through a reseller (referred to as he) on a platform, as demonstrated by the motivational examples in Section 1. The platform-based resell channel is more cost-efficient in the sales process than the manufacturer's direct channel (Niu et al., 2021; Sun et al., 2021; Nie et al., 2022). Therefore, we normalize the reseller's unit selling cost in the platform to0, and denote the manufacturer's unit selling cost in the direct channel by$d$. The value of d captures the cost difference between the two channels. However, for each product sold in the resell channel, the reseller pays the platform a fraction α of his revenue as the fee for accessing the consumers (Abhishek et al., 2016; Yang and Tang, 2019; He et al., 2021). To be consistent with reality, we assume $\alpha < 0.2$ (e.g., Amazon charges 6% per-sale commission for personal computers and 8% for cell phones).

To model anti-counterfeiting in the platform, we make the following assumptions. First, the reseller has an incentive to sell a fake product. The fake product is produced by the reseller himself or purchased from a third party at a constant wholesale price. The fake product has a unit production cost lower than c (Yi et al., 2020). For analytical transparency, the unit cost for the reseller to acquire the fake product is normalized to0. Thus, the value of c essentially measures the cost advantage of the fake product. Second, in the absence of anti-counterfeiting, the fake products accounts for a proportion ϕ of all products sold by the reseller in the platform. The value of ϕ indicates the availability of fake products in the marketplace, and it depends highly on intellectual property law enforcement (Qian, 2014; Qian et al., 2015; Pun et al., 2021). The level of law enforcement in a specific market is usually common knowledge, and hence the parameter ϕ is assumed to be public information. Specifically, in a market with strict law enforcement, ϕ is small; vice versa. Third, the platform and manufacturer can adopt anti-counterfeiting technology to eliminate fake products. For example, the platform and manufacturer can use blockchain technology to provide digital authentication for each product, which will help consumers identify the authenticity of the product, making fake products unprofitable (Choi, 2019; Shen et al., 2022). Therefore, in the presence of anti-counterfeiting, the value of ϕ can be reduced to0. We notice that the authentic manufacturers may be involved in the sales of fake products, which, however, is not common in practice. In this study, following the relevant literature on anti-counterfeiting, e.g., Qian et al. (2015), Gao (2018), Pun et al. (2021), we focus on the context where the manufacturer never sells fake products to consumers.

Following the relevant literature on channel competition, e.g., Jerath et al. (2010), Hu et al. (2016), and Li (2021), we assume that consumers have different preferences between the direct channel and the resell channel. Without loss of generality, a continuum of consumers is uniformly spread on a Hotelling line over the interval $[0, 1]$ with density1, and the two channels are located at each end of the horizontal line. The channel preference of every consumer is determined by her/his location $x \in [0, 1]$ on the line. One consumer places a valuation v for the authentic product produced by the manufacturer and a valuation 0 for the fake product. Here, the value of v represents the valuation difference between the authentic product and the fake product. Therefore, an ordinary consumer's valuation on the product sold in the platform is$(1 - \phi)v$. In other words, because of the existence of fake products on the platform, consuemrs have a lower valuation on the product sold by the reseller than the product sold directly by the manufacturer.

Moreover, each channel can provide services that are valued by consumers, e.g., extended warranty, installment payment, and home delivery. For the sake of clarity, we normalize consumers' valuation on the services provided by the direct channel to0, and denote the valuation on the services in the resell channel byδ. In practice, different channels usually have different efficiencies in providing services. Thus, even with the same service cost, different channels usually have different services. Without loss of generality, we normalize the service cost to 0 (Song et al., 2021; Zhao et al., 2021; Bae et al., 2022). It is worth noting that the effect of a positive service cost can be reflected by the existing parameter$d$. That is, if the direct channel has a disadvantage in the service cost, the disadvantage can be captured by a higher value of$d$, vice versa. Therefore, the assumption of a zero service cost is not so harmful as it looks. The parameter δ captures the difference between the two channels' services. Specifically, $\delta > 0$ ($\delta < 0$) indicates that the platform provides better (worse) services than the manufacturer.

Therefore, a typical consumer located at x acquires a net consumption utility of $v - tx - p_m$ buying the product directly from the manufacturer at a price ofp_m, or $(1 - \phi)v + \delta - t(1 - x) - p_r$ buying the product from the reseller on the platform at a price ofp_r, where the parameter t measures the strength of channel preference. In this study, we focus on the case of deceptive counterfeits (Pun et al., 2021); that is, the fake product has the same price as the authentic product, and hence no consumer can distinguish them through their prices. To ensure the market is completely covered, we assume that v is sufficiently large, that is$v > t_1^v$, so that each consumer have a positive net consumption utility and buys one product in equilibrium. In other words, t_1^v is the threshold of$v$, ensuring that all customers

have a positive utility. Meanwhile, to focus on the competition between the two channels, we consider that the cost disadvantage of the direct channel is moderate, that is $t_1^d < d < t_2^d$, so that the demand of each channel is positive. Specifically, t_1^d (t_2^d) is the threshold of d that ensures a positive demand for the resell (direct) channel. The mathematical expressions of t_1^v,t_1^d, and t_2^d are provided in the Appendix. Fig. 1 illustrates the channel structure.

Let $\tilde{x}$ indicate the location of the marginal customer who is indifferent between buying the product directly from the manufacturer and buying the product from the reseller on the platform. Given p_m and p_r, we can derive the demand functions for the direct channel and the resell channel by solving $v - t\tilde{x} - p_m = (1-\phi)v + \delta - t(1-\tilde{x}) - p_r$, as follows:

$$D_m = \tilde{x} = \frac{\phi v - \delta + t - p_m + p_r}{2t} \tag{1}$$

$$D_r = 1 - \tilde{x} = \frac{\delta - \phi v + t - p_r + p_m}{2t} \tag{2}$$

Events happen in a sequence in the model. First, the platform and the manufacturer make a strategic decision of whether to use the anti-counterfeiting technology. The investment cost of the technology is F. By comparing the revenues with and without the anti-counterfeiting technology, we can obtain the value of the technology and straightforwardly, the platform and the manufacturer will invest in the technology if and only if the value outweighs the cost. In this sense, the decision on anti-counterfeiting is endogenous, and the investment in anti-counterfeiting is equivalent to the ability of reducing the proportion of fake products. Second, the manufacturer and the reseller engage in a Stackelberg competition. The manufacturer charges w to the reseller and p_m to consumers in the direct channel as the leader; next, the reseller as the follower decides on the retail price p_r in the resell channel. The assumption that the manufacturer plays a leading role in the interaction with the reseller is widely practiced in many industries and is also commonly used in the relevant literature, see, e.g., Karp and Perloff (2005), Su (2010), and Cui et al. (2014). Finally, consumers make their purchase decisions, and firms obtain their revenues. We assume that all firms are risk-neutral and profit-seeking. For the convenience of analysis, we omit the consideration of the investment cost of the anti-counterfeiting technology before we obtain the revenues of anti-counterfeiting for the platform and the manufacturer (Choi, 2019). Regardless of the investment cost of the anti-counterfeiting technology, the revenues of the manufacturer, the reseller, and the platform are.

$$\pi_m = (w - c)(1 - \phi)D_r + (p_m - c - d)D_m \tag{3}$$

$$\pi_r = ((1 - \alpha)p_r - (1 - \phi)w)D_r \tag{4}$$

$$\pi_p = \alpha p_r D_r \tag{5}$$

For ease of reference, all notations are summarized in Table 1.

4. Analysis

We first investigate the interaction between the manufacturer and the reseller by taking the anti-counterfeiting strategy as given. Based on sub-game equilibria in the cases with and without anti-counterfeiting, we acquire the revenues of the manufacturer, the reseller, and the platform. Finally, a comparison of these revenues in equilibrium reveals the implications of anti-counterfeiting, and then we analyze who is willing to incur the cost of anti-counterfeiting technology; that is, the incentives for the platform and the

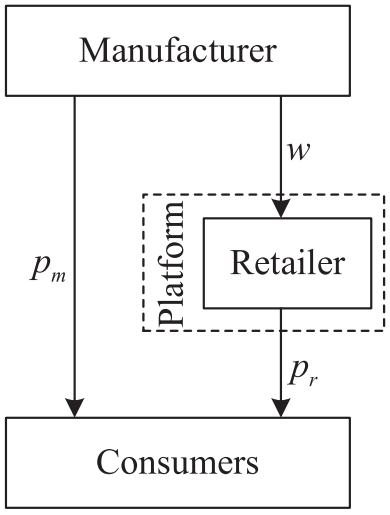


Fig. 1. The channel structure.

Table 1
Notation.

Symbol	Definition
Parameters	
c	Unit production cost of the product
d	Unit selling cost in the direct channel
F	The investment cost of the anti-counterfeiting technology
v	Consumers' valuation on the authentic product
δ	Consumers' valuation on the platform's services
t	Strength of consumers' channel preference
ϕ	Proportion of the fake product on the platform,$\phi \in [0,1]$
α	Commission charged by the platform,$\alpha \in [0,1]$
Decision and auxiliary variables	
D_m	Demand for the manufacturer's direct channel
D_r	Demand for the resell channel
w	Wholesale price to the reseller
p_m	Retail price of the manufacturer in the direct channel
p_r	Retail price of the reseller in the platform
π_i	Revenue of$i,i \in \{m,r,p\}$

manufacturer to invest in anti-counterfeiting technology.

4.1. The case without anti-counterfeiting

In the case without anti-counterfeiting, denoted by a superscript$N$, a proportion ϕ of products sold by the reseller on the platform are fake products. The revenues of the manufacturer and the reseller are shown in Equations (3) and (4). We use backward induction to solve for the equilibrium outcome. Taking the wholesale price as given, we can obtain the optimal retail price responses of the manufacturer and the reseller, as follows.

Lemma 1. *In the case without anti-counterfeiting, for a given wholesale price$w$, the optimal retail price responses of the manufacturer and the reseller are*$p_m^{N^*}(w) = \frac{(1-\alpha)((c+v)\phi+d+3t-\delta)+(2-\alpha)(1-\phi)w}{2(1-\alpha)}$,$p_r^{N^*}(w) = \frac{(1-\alpha)((c-v)\phi+d+5t+\delta)+(4-\alpha)(1-\phi)w}{4(1-\alpha)}$.

It is clearly shown in Lemma 1 that the optimal retail price responses of the manufacturer and the reseller both increase in the wholesale price. On one hand, for a higher wholesale price, it implies that the reseller incurs a higher cost to purchase the product, and hence it is the intuition for the reseller to set a higher retail price on the platform. On the other hand, a higher wholesale price indicates that the manufacturer can obtain a higher profit margin in the resell channel, and thus she has an incentive to increase (decrease) the demand for the resell (direct) channel. To this end, the manufacturer increases the retail price in the direct channel. We also find that as the wholesale price increases, the reseller increases the retail price on the platform faster than the manufacturer in the direct channel because of the well-known effect of double marginalization.

Anticipating the retail price responses, the manufacturer sets the wholesale price that maximizes her revenue.

Proposition 1. *In the case without anti-counterfeiting, the manufacturer's optimal wholesale price is*$w^{N^*} = \frac{(1-\alpha)(8v-3\phi c-5\phi v-3d+5\delta-15t)}{(1-\phi)(8-3\alpha)}$.

It can be seen from Proposition 1 that as consumers' valuation on the product increases, the manufacturer can set a higher wholesale price. This indicates that if the manufacturer's product has a high valuation, the manufacturer will have more power to deal with the reseller, being able to set a higher wholesale price. We also find that as consumers' valuation on the platform's services increases, the manufacturer's wholesale price will increase. This is because the reseller can set a higher retail price if the platform can provide better services, allowing the manufacturer to set a higher wholesale price. Additionally, we also note that as the proportion of fake products increases, the wholesale price will rise. This is in line with intuition. Because the proportion of fake products increases, the proportion of genuine products that the reseller needs to purchase from the manufacturer decreases, thereby reducing the unit resell cost, so the reseller is willing to pay a higher wholesale price.

Based on the optimal wholesale price, we can obtain the optimal retail prices and the demands, which are provided in the Appendix. The following result characterizes how the retail prices and the demands are shaped by the consumers' valuation of the platform's services (δ), and the proportion of the fake product on the platform (ϕ). In this paper, the signs0,−, and + denote no change, a decrease, and an increase in equilibrium with respect to an incremental change in the corresponding parameter, respectively. To

Table 2
The impacts of δ and ϕ on decisions.

Parameter	$p_m^{N^*}$	$p_r^{N^*}$	$D_m^{N^*}$	$D_r^{N^*}$
δ	+	+	−	+
ϕ	$\pm^{\bullet}$	−	$\pm^{\bullet\bullet}$	$\pm^{\bullet}$

$^{\cdot}$+ *if*$v \leqslant t_2^v$; − *otherwise.* $^{\cdot\cdot}$+ *if*$v > t_2^v$; − *otherwise.*

facilitate the analysis, we define four thresholds$t_2^v, t_1^\delta, t_1^\phi$ and t_2^ϕ (their expressions are provided in the Appendix).

Firstly, Table 2 indicates that as consumers' valuation on platform services increases, the reseller's retail price increases, which is in line with intuition. And a higher retail price of reseller makes the manufacturer to set higher wholesale price. Which indicates that the manufacturer can obtain a higher profit margin in the resell channel, and thus she will increase the demand for the resell channel while decrease the demand for the direct channel by increasing the retail price in the direct channel.

Second, Table 2 also reveals the different effects of the proportion of fake products on the retail prices and the demands. Fig. 2(a) shows that with the increase of the proportion of the fake product, consumers' valuation of platform products decreases, which makes the retail price that the reseller can set decreased. When the product valuation is large, the manufacturer can set a high wholesale price, but the wholesale loss of the manufacturer from resell channel caused by the fake product is also large. Hence, as shown by the dotted line in Fig. 2(b), Fig. 2(c), and Fig. 2(d), with the increase in the proportion of the fake product, the manufacturer will set a lower retail price to decrease the demand for the resell channel while increase the demand for the direct channel to reduce the wholesale loss from the resell channel. However, when the product valuation is small, on one hand, the manufacturer can only set a low wholesale price, so the wholesale loss from the resell channel caused by the fake product is small. On the other hand, as the proportion of the fake product increases, the wholesale price will rise, so the manufacturer can obtain a higher profit margin in the resell channel. Hence, as shown by the dashed line in Fig. 2(b), Fig. 2(c), and Fig. 2(d), with the increase of the proportion of the fake product, the manufacturer will set a higher retail price to increase the demand for the resell channel while decrease the demand for the direct channel.

Based on the optimal wholesale price, the optimal retail prices, and the demands, we can obtain the revenues of the manufacturer, the reseller, and the platform. The expressions of their revenues are provided in the Appendix. We now examine how the revenues are shaped by the consumers' valuation on the platform's services (δ), and the proportion of the fake product on the platform (ϕ). The

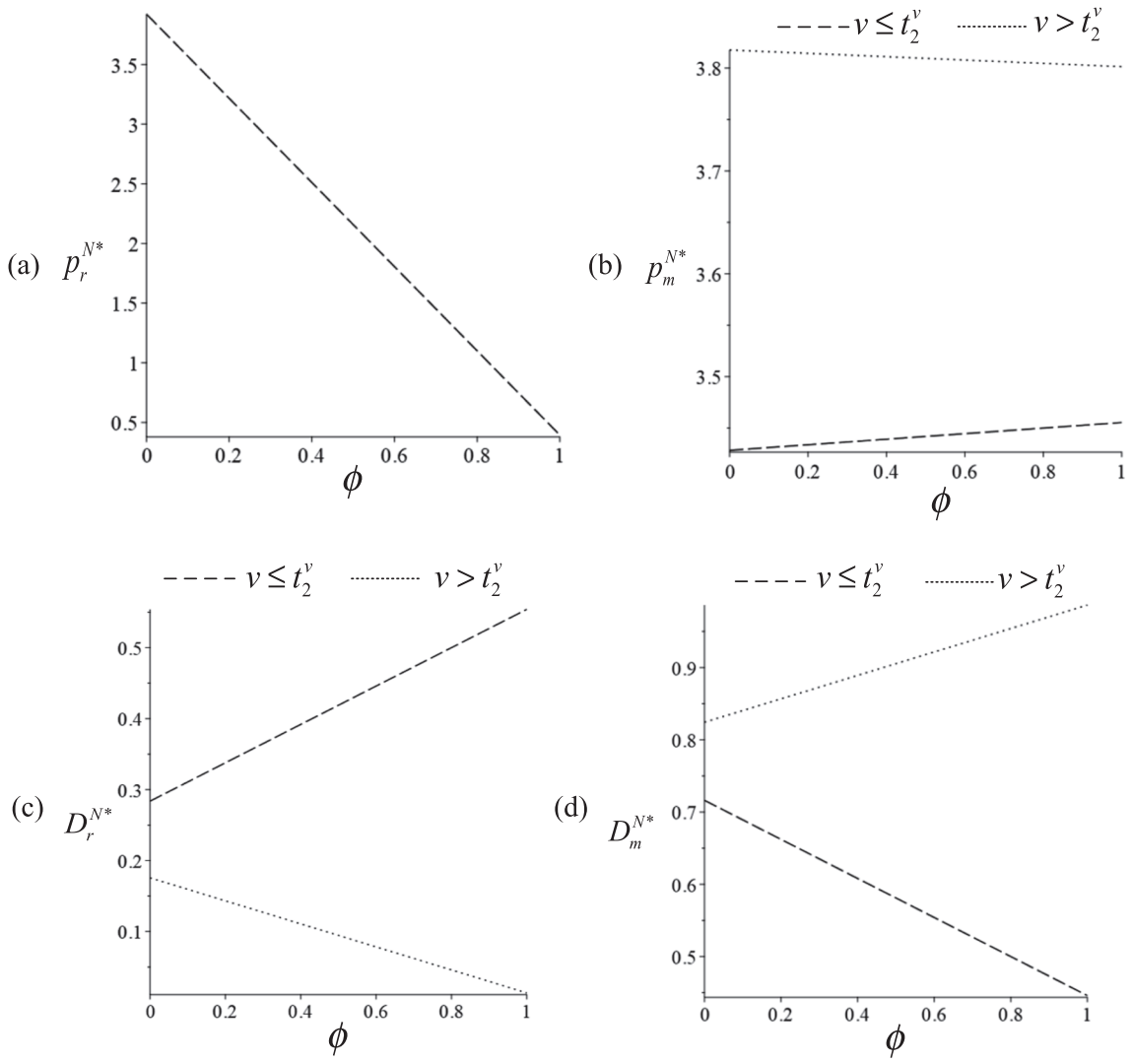


Fig. 2. Impacts of the proportion of the fake product on decisions.

impacts of parameters are presented in Table 3.

Firstly, Table 3 indicates that as consumers' valuation on the platform's services increases, the revenues of both the platform and the reseller increase, which is consistent with the intuition for the increases of the reseller's retail price and the demand for the resell channel. However, as the valuation of platform's services increases, the revenue of the manufacturer will increase only when the valuation of platform's services is large. The reason is that, only when the valuation of platform's services is large, the increase in wholesale price caused by the increase in platform service valuation will offset the decline in the demand for the direct channel.

Second, Table 3 also reveals the non-monotonic impact of the proportion of the fake product on revenues. For the manufacturer, with the proportion of the fake product increases, the wholesale price will increase, while either the retail price or the demand for the direct channel will decrease. Only when the proportion of the fake product is high, the increase in the wholesale price can offset the decrease in the demand or the retail price for the direct channel. Hence, as shown by the dotted line in Fig. 3(a), with the proportion of the fake product increases, the manufacturer's revenue increases (decreases) when the proportion of the fake product is high (low). For the reseller, as shown by the dotted line in Fig. 3(b), when the product valuation is high, with the increase of the proportion of the fake product, both the reseller's retail price and the market share decrease, making his revenue declined. As shown by the dashed line in Fig. 3(b), when the product valuation is low, with the increase of the proportion of the fake product, the reseller's market share increases and plays a major role, making his revenue rised.

For the platform, only when the product valuation is low and the proportion of the fake product is small, with the increase of the proportion of the fake product, reseller's retail price decreases slightly, which can be offset by the increase of the demand for the resell channel. Hence, as shown by the dashed line in Fig. 3(c), only when the product valuation is low and the proportion of the fake product is small, the platform's revenue will rise with the increase of the proportion of the fake product.

4.2. The case with anti-counterfeiting

In the case with anti-counterfeiting, denoted by a superscript$A$, technologies such as blockchain are used to fight counterfeits. Because blockchain technology can accurately identify the authenticity of the product, the reseller no longer sells the fake product. With an analysis process similar to the case without anti-counterfeiting, we obtain results in the case with anti-counterfeiting. The specific expressions can be obtained by replacing the proportion of the fake product ϕ by 0 in the equilibrium results in the case without anti-counterfeiting. Since the equilibrium expressions in the case with anti-counterfeiting are similar to that in the case without anti-counterfeiting, these results are omitted in this subsection for the sake of brevity. We now examine how the consumers' valuation on the platform's services (δ) shape the equilibrium. The impacts of parameter are presented in Table 4 (the expression of t_2^{δ} is provided in the Appendix).

Compared with Table 2, Table 4 indicates that, in the case with anti-counterfeiting, the consumers' valuation on the platform's services has a similar impact on the equilibrium as it does in the case without anti-counterfeiting.

4.3. The effect of anti-counterfeiting and investment incentives

In this subsection, first, without considering the cost of anti-counterfeiting technology, we will analyze the effect of anti-counterfeiting by comparing the equilibrium results in the two cases with and without anti-counterfeiting. Then, by comparing the revenue and the cost of anti-counterfeiting technology, we examine the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology. To facilitate analysis, we define three thresholds$t_p^{\delta}, t_p^{\phi}, t_m^{\phi}$, and their expressions are provided in the Appendix.

We compare the wholesale prices of the platform in the two cases with and without anti-counterfeiting, to analyze the impact of anti-counterfeiting on the wholesale price. As shown in Fig. 4, anti-counterfeiting has reduced the wholesale price, i.e.,$w^{A^*} < w^{N^*}$. This is because anti-counterfeiting makes the reseller to purchase all products from the manufacturer, resulting in a lower incentive for reselling. So, the manufacturer needs to motivate the reseller by reducing the wholesale price. This also suggests that if fake products can be sold, the reseller is willing to pay a higher wholesale price to the manufacturer. This is a potential benefit of fake products for the manufacturer because the reseller's willingness to pay a higher wholesale price means a higher marginal profit of the resell channel.

Next, we compare the retail prices and the demands of the reseller and the manufacturer in the two cases with and without anti-counterfeiting, to analyze the impact of anti-counterfeiting on the retail prices and the demands.

Proposition 2. *The impact of anti-counterfeiting on the retail prices and the demands.*

Table 3
The impacts of δ and ϕ on the revenues.

Parameter	$\pi_m^{N^*}$	$\pi_r^{N^*}$	$\pi_p^{N^*}$
δ	$\pm^{*}$	$+$	$+$
ϕ	$\pm^{\bullet}$	$\pm^{**}$	$\pm^{\bullet\bullet}$

$^{*}+ if \delta > t_1^{\delta}; - otherwise.$ $^{\cdot}+ if \phi > t_1^{\phi}; - otherwise.$

$^{**}+ if v < t_2^{v}; - otherwise.$ $\cdots+ if\ v < t_2^{v}\ and \phi < t_2^{\phi}; - otherwise.$

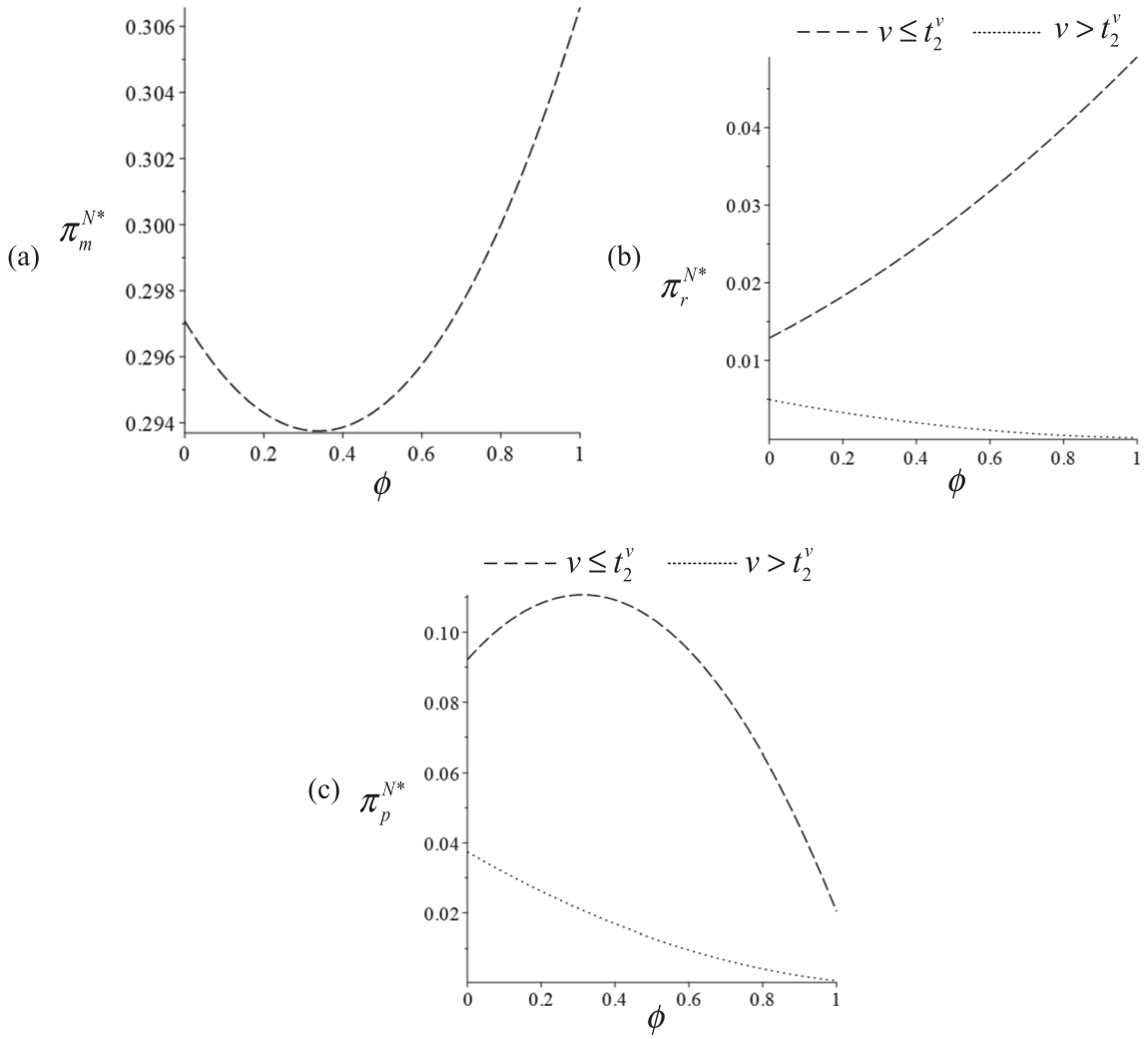


Fig. 3. Impacts of proportion of the fake product on the revenues.

Table 4
The impacts of δ on decisions.

Parameter	p_m^{A*}	p_r^{A*}	D_m^{A*}	D_r^{A*}	π_m^{A*}	π_r^{A*}	π_p^{A*}
δ	+	+	−	+	$\pm^*$	+	+

*+ *if* $\delta > t_2^\delta$; − *otherwise.*

(i) *anti-counterfeiting increases the retail price of the reseller;*
(ii) *if* $v > t_2^v$, *anti-counterfeiting increases the retail price of the manufacturer and the demand of the reseller and decreases the demand of the manufacturer; otherwise, anti-counterfeiting decreases the retail price of the manufacturer and the demand of the reseller and increases the demand of the manufacturer.*

Proposition 2 highlights the impact of anti-counterfeiting on the retail prices and demands of the manufacturer and the reseller. For the reseller, the direct impact of anti-counterfeiting is to increase the valuation on the product sold by the reseller so that consumers are willing to pay a higher price. So, as shown in Fig. 5(a), anti-counterfeiting increases the retail price of the reseller, i.e., $p_r^{A*} > p_r^{N*}$.

For the manufacturer, anti-counterfeiting has both positive and negative effects. The positive impact of anti-counterfeiting is to avoid the wholesale loss caused by fake product in resell channel. The negative impact of anti-counterfeiting is to lower the wholesale price. When consumers' valuation on the product is high, i.e., $v > t_2^v$, the manufacturer can set a high wholesale price, and the wholesale loss caused by fake product in the resell channel is correspondingly large. In this scenario, anti-counterfeiting allows the

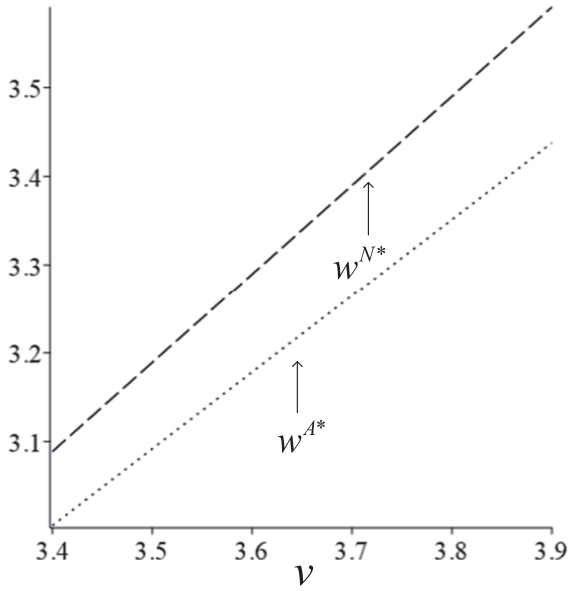


Fig. 4. The impact of anti-counterfeiting on the optimal wholesale price.

manufacturer to avoid the large wholesale loss from the fake product, offsetting the decline in the wholesale price. So, as shown in Fig. 5(b), Fig. 5(c), and Fig. 5(d), when consumers' valuation on the product is high, anti-counterfeiting makes the manufacturer to increase the demand for the resell channel and decrease the demand for the direct channel by raising the retail price in the direct channel. On the contrary, when consumers' valuation on the product is low, i.e.,$v \leqslant t_2^v$, the manufacturer can only set a low wholesale price, and the wholesale loss caused by the fake product is also small. In this scenario, anti-counterfeiting only allows the manufacturer to avoid a small wholesale loss which is not enough to offset the decline in wholesale price. So, as shown in Fig. 5(b), Fig. 5(c), and Fig. 5(d), when consumers' valuation on the product is low, anti-counterfeiting makes the manufacturer to decrease the demand for the resell channel and increase the demand for the direct channel by lowering the retail price in the direct channel.

Next, we will analyze the anti-counterfeiting revenue of the platform and the manufacturer separately by comparing their revenues in the two cases with anti-counterfeiting and without anti-counterfeiting.

Proposition 3. *The anti-counterfeiting revenue of the platform.*

(i) *if* $\phi > t_p^\phi$ *or* $v > t_2^v$, *the anti-counterfeiting revenue of the platform is positive*;

(ii) *otherwise, the anti-counterfeiting revenue of the platform is negative.*

From Proposition 3 and Fig. 6, we have the following findings. First, when the proportion of the fake product is high, or consumers' valuation on the product is high, i.e., $\phi > t_p^\phi$ or$v > t_2^v$, corresponding to the region (i) of Fig. 6, the anti-counterfeiting revenue of the platform is positive. This is because when the proportion of the fake product is high, or when consumers' valuation on the product is high, anti-counterfeiting can significantly increase consumers' valuation on the product sold in the platform. Thus, the reseller can massively increase the retail price. Although the reseller pays a fixed fraction α of its revenue to the platform, the platform can benefit from anti-counterfeiting because of the larger revenue when the reseller strategically raises the retail price in the platform. In reality, the products on the JD platform are more valuable than Pinduoduo; that is, the valuation is larger, so JD will obtain a positive revenue from anti-counterfeiting and take anti-counterfeiting action more often. On the Pinduoduo platform, the proportion of fake products is usually large, so Pinduoduo may also obtain a positive revenue from choosing anti-counterfeiting and taking anti-counterfeiting action.

Second, when the proportion of fake products and consumers' valuation on the product are both low, i.e., $\phi \leqslant t_p^\phi$ and$v \leqslant t_2^v$, corresponding to region (ii) in Fig. 6, the anti-counterfeiting revenue of the platform is negative. This is because when the proportion of the fake product and consumers' valuation on the product are both low, anti-counterfeiting will not bring enough increase in consumers' valuation on the platform products. Even if the retail price of the reseller increases, the competition from the manufacturer makes the sales of the reseller drop a lot, resulting in the commissions charged by the platform decreased instead. This implies that the platform may not be willing to completely eliminate fake products. For example, as Pinduoduo, even though there have been many reports exposing its counterfeit problems, the platform still has not adopted strong anti-counterfeiting measures to completely eliminate counterfeit products.

Third, it is worth noting that when$\delta > t_p^\delta$, $t_p^\phi < 0$ always holds and satisfies$\phi > t_p^\phi$. This means that when consumers' valuation of the platform's services is high, the platform can also obtain a positive revenue from anti-counterfeiting. At the same time, in Section 4.2,

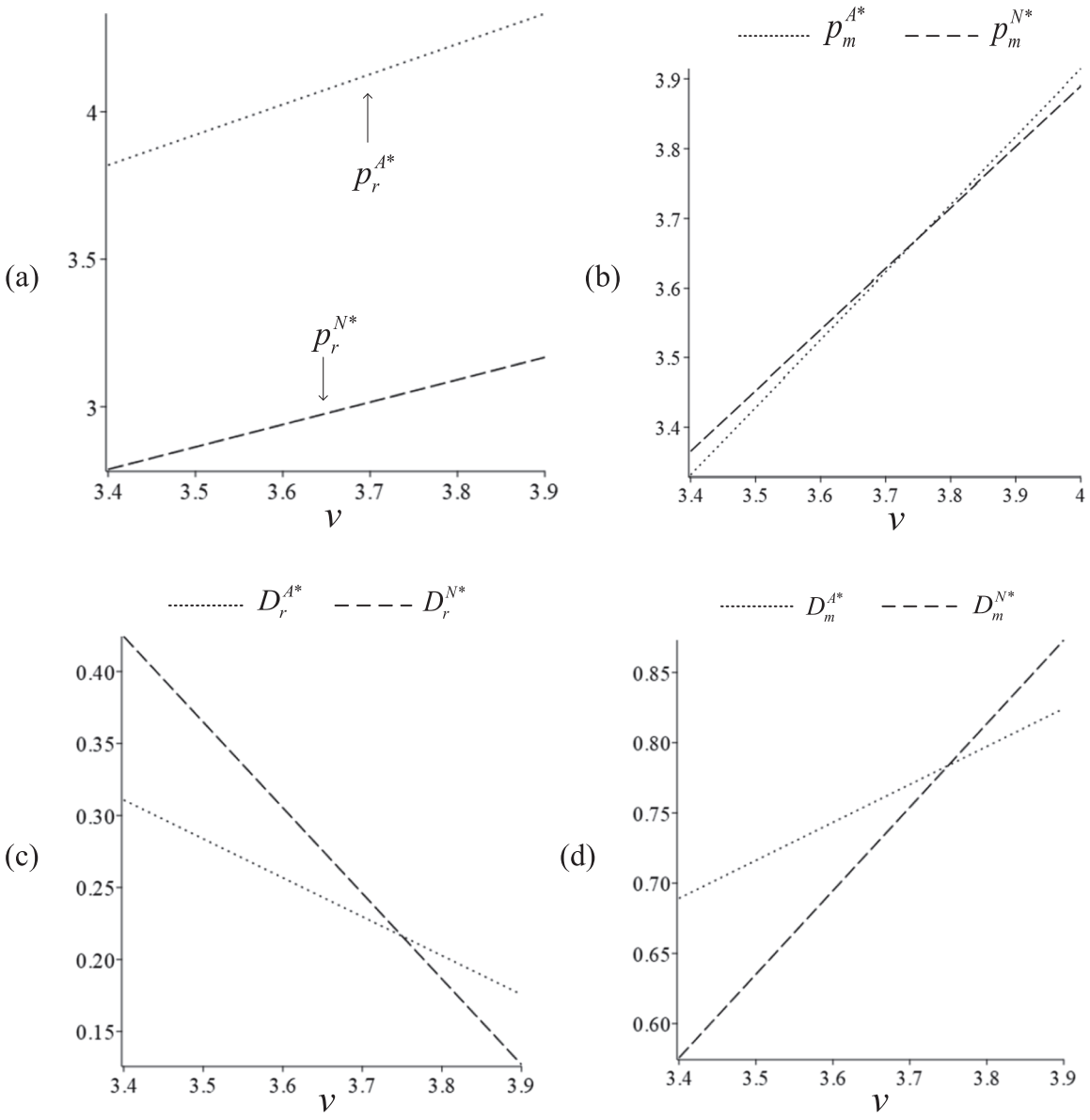


Fig. 5. The impact of anti-counterfeiting on the retail prices and demands.

our analysis indicates that the revenue of the platform increases with the increase in the consumers' valuation on the platform services in the case with anti-counterfeiting. Therefore, when anti-counterfeiting technology is used to drive fake products out of the market, in order to obtain greater revenue, the platform should provide better platform services than before. For example, when Taobao was facing strict supervision of counterfeit, Alibaba launched Tmall and provided better platform services. Meanwhile, compared with Taobao, Tmall is also more aggressive and more stringent in anti-counterfeiting measures.

Proposition 4. *The anti-counterfeiting revenue of the manufacturer.*

(i) *if* $\phi < t_m^{\phi}$, *the anti-counterfeiting revenue of the manufacturer is positive*;

(ii) *otherwise*, *the anti-counterfeiting revenue of the manufacturer is negative.*

Proposition 4 shows that when the proportion of the fake product on the platform is low, i.e., $\phi < t_m^{\phi}$, corresponding to the region in Fig. 7 (i), the anti-counterfeiting revenue of the manufacturer is positive. However, when the proportion of the fake product on the platform is high, i.e., $\phi \geq t_m^{\phi}$, corresponding to the region (ii) in Fig. 7, the anti-counterfeiting revenue of the manufacturer is negative. The reasons are explained as follows. On the one hand, anti-counterfeiting increases either the price or the demand in the direct channel. On the other hand, the wholesale price will rise as the proportion of fake products increases, and anti-counterfeiting will

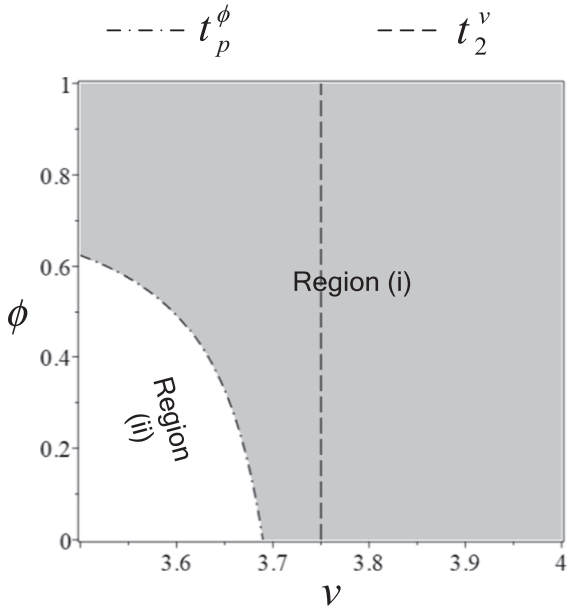


Fig. 6. The impact of anti-counterfeiting on the revenue of the platform.

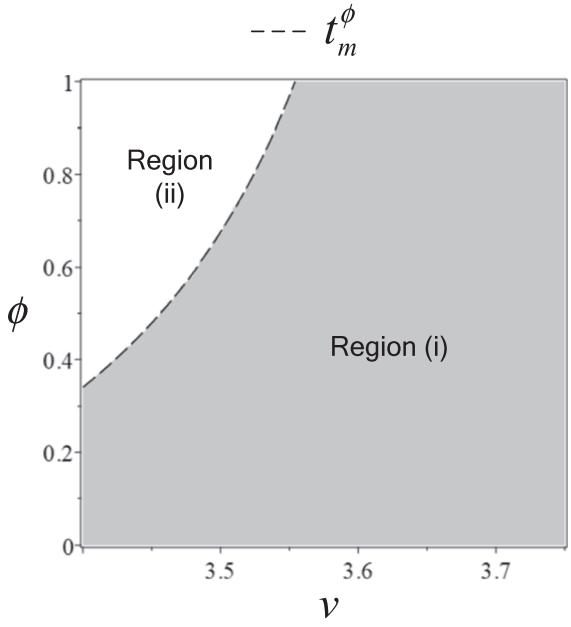


Fig. 7. The impact of anti-counterfeiting on the revenue of the manufacturer.

lower the wholesale price. When the proportion of the fake product on the platform is low, the manufacturer can only set a small wholesale price. In this scenario, anti-counterfeiting makes the wholesale price decrease slightly that will not offset the increase of the price or the demand in the direct channel, so the anti-counterfeiting revenue of the manufacturer is positive. On the contrary, when the proportion of the fake product on the platform is high, the manufacturer can set a high wholesale price. In this scenario, anti-counterfeiting makes the wholesale price decreased sharply, offsetting the increase of the price or the demand in the direct channel, so the anti-counterfeiting revenue of the manufacturer is negative.

Proposition 4 also indicates that the manufacturer will not crack down on the fake product if the proportion of the fake product is high. For example, Pinduoduo has a relatively high number of fake products, but the manufacturers that supply Pinduoduo are not very active in anti-counterfeiting. By contrast, there are relatively fewer fake products on JD.com, but the manufacturers are more aggressive in anti-counterfeiting.

Now, by comparing the revenue and cost of anti-counterfeiting technology, we analyze who is willing to incur the cost of anti-counterfeiting technology, the platform, or the manufacturer. That is, the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology. For the convenience of analysis, we define two thresholds t_m^F andt_p^F, and their expressions are shown in the Appendix.

Proposition 5. *The incentives for the platform and manufacturer to invest in anti-counterfeiting technology.*

(i) *if*$F \leqslant \min\left\{t_m^F, t_p^F\right\}$, *both the platform and the manufacturer have incentives to invest in anti-counterfeiting technology;*

(ii) *if* $\min\left\{t_m^F, t_p^F\right\} < F \leqslant \max\left\{t_m^F, t_p^F\right\}$ *and*$\delta \leqslant t_3^\delta$, *only the manufacturer has the incentive to invest in anti-counterfeiting technology;*

(iii) *if* $\min\left\{t_m^F, t_p^F\right\} < F \leqslant \max\left\{t_m^F, t_p^F\right\}$ *and*$\delta > t_3^\delta$, *only the platform has the incentive to invest in anti-counterfeiting technology;*

(iv) *otherwise, neither the platform nor the manufacturer has the incentive to invest in anti-counterfeiting technology.*

The incentives for the platform and the manufacturer to invest in anti-counterfeiting technology are graphically illustrated in Fig. 8. As shown in region (i) of Fig. 8, if the investment cost of anti-counterfeiting is low, i.e.,$F \leqslant \min\left\{t_m^F, t_p^F\right\}$, both the platform and the manufacturer can afford the cost, so both have incentives to invest in anti-counterfeiting technology.

However, as shown in the region (ii) and region (iii) of Fig. 8, if the investment cost of anti-counterfeiting is relatively high, i.e.,$\min\left\{t_m^F, t_p^F\right\} < F \leqslant \max\left\{t_m^F, t_p^F\right\}$, only the manufacturer has the incentive to invest in anti-counterfeiting technology when the consumers' valuation on the platform's services is small, i.e.,$\delta \leqslant t_3^\delta$, while only the platform has the incentive to invest in anti-counterfeiting technology when the consumers' valuation on the platform's services is large, i.e.,$\delta > t_3^\delta$. The reasons are explained as follows. On the one hand, anti-counterfeiting will increase the retail price of the reseller in the platform. On the other hand, the larger the consumers' valuation on the platform's services, the more consumers are willing to buy the product sold by the reseller. When the consumers' valuation on the platform's services is large, there are many consumers buying the product sold by the reseller, so the platform has more incentive to crack down on counterfeiting to make those consumers to pay a higher retail price. On the contrary, when the consumers' valuation on the platform's services is small, not many consumers will buy the product sold by the reseller. In this scenario, anti-counterfeiting increases the retail price of the reseller but may further discourage consumers from buying the product sold by the reseller, which makes the platform has less incentive to invest in anti-counterfeiting technology than the manufacturer.

If the investment cost of anti-counterfeiting is very high, as shown in the region (iv) of Fig. 8, i.e.,$F > \max\left\{t_m^F, t_p^F\right\}$, it is clear that since neither the platform nor the manufacturer can afford the cost of anti-counterfeiting technology, they have no incentive to invest in anti-counterfeiting technology.

Next, we analyze the impact of anti-counterfeiting on consumer surplus and social welfare. We denote consumer surplus and social welfare as *CS* and *SW* respectively, where$CS = \int_0^{Dm} (v - xt - p_m)\, dx + \int_{Dm}^1 ((1-\phi)v + \delta - (1-x)t - p_r)dx$; and$SW = \pi_p + \pi_m + \pi_r + CS$.

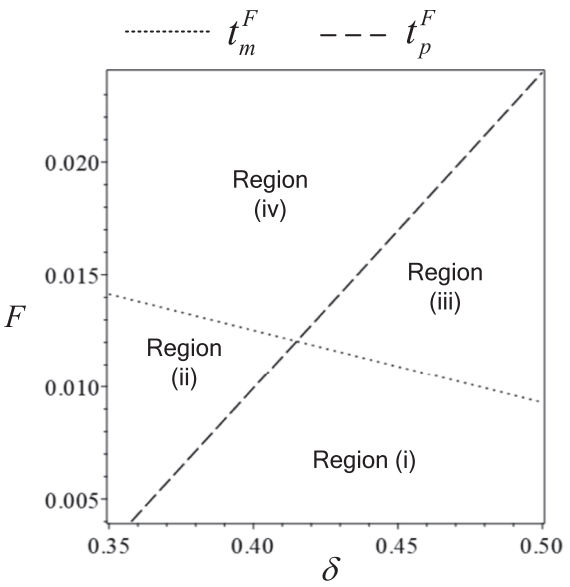


Fig. 8. The investment incentives of the manufacturer and platform in the main model.

Proposition 6. *The impact of anti-counterfeiting on consumer surplus:*

(i) *if* $\phi < t_c^{\phi}$, *the impact of anti-counterfeiting on consumer surplus is positive*;

(ii) *otherwise, the impact of anti-counterfeiting on consumer surplus is negative.*

Proposition 6 shows that when the proportion of the fake product on the platform is low, i.e.,$\phi < t_c^{\phi}$, corresponding to the region in Fig. 9 (i), the impact of anti-counterfeiting on consumer surplus is positive. However, when the proportion of the fake product on the platform is high, i.e.,$\phi \geqslant t_c^{\phi}$, corresponding to the region (ii) in Fig. 9, the impact of anti-counterfeiting on consumer surplus is negative. The reasons are explained as follows. On the one hand, cracking down on the fake product increases the valuation on the product sold by the reseller. On the other hand, the retail price of the reseller decreases as the proportion of the fake product on the platform increases, while cracking down on the fake product increases the retail price of reseller. When the proportion of the fake product is low, the retail price of the reseller is relatively high. Cracking down on the fake product increases the retail price of reseller slightly, which will not offset the increase in the valuation on the product sold by the reseller. However, when the proportion of the fake product is high, the retail price of the reseller is relatively low, and cracking down on the fake product will increase the retail price of reseller sharply, offsetting the increase in the valuation on the product sold by the reseller.

Proposition 7. *The impact of anti-counterfeiting on social welfare.*

(i) *if* $\phi > t_s^{\phi}$ *or* $v > t_2^{v}$, *the impact of anti-counterfeiting on social welfare is positive*;

(ii) *otherwise, the impact of anti-counterfeiting on social welfare is negative.*

From Proposition 7 and Fig. 10, we have the following findings. First, when the proportion of the fake product is high, or consumers' valuation on the product is high, i.e., $\phi > t_s^{\phi}$ or$v > t_2^{v}$, corresponding to the region (i) of Fig. 10, the impact of anti-counterfeiting on social welfare is positive. This is because, on the one hand, when the proportion of the fake product is high, or consumers' valuation on the product is high, cracking down on fake product makes the manufacturer avoid a large wholesale loss caused by the fake product; On the other hand, cracking down on fake product increases the valuation on the product sold by the reseller and thus increase their consumption utility, while allowing the reseller to set a higher retail price and the platform will earn more commissions. Second, when the proportion of fake products and consumers' valuation on the product are both low, i.e., $\phi \leqslant t_s^{\phi}$ and$v \leqslant t_2^{v}$, corresponding to region (ii) in Fig. 10, the impact of anti-counterfeiting on social welfare is negative. This is because, when the proportion of the fake product and the consumers' valuation on the product are both low, cracking down on fake product only makes the manufacturer avoid a small wholesale loss, and the increase in the valuation on the product sold by the reseller is also small, which is not enough to offset the negative impact of the reseller raising retail price.

5. Extension

In this extension, we investigate the impact of network externality to check the robustness of the results. Specifically, we consider

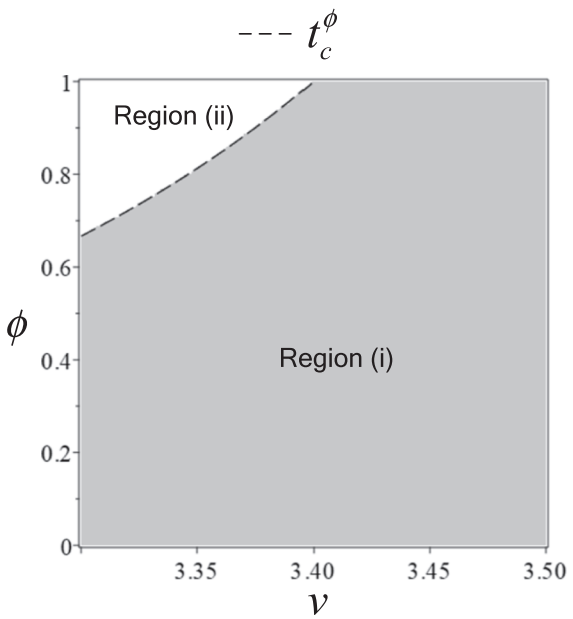


Fig. 9. The impact of anti-counterfeiting on consumer surplus.

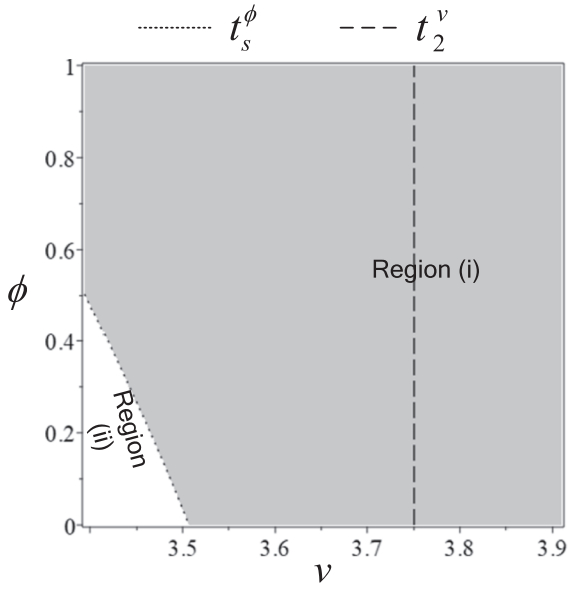


Fig. 10. The impact of anti-counterfeiting on social welfare.

the negative externality of counterfeits on the reputation of the platform, that is, the more counterfeits sold on the platform, the worse the reputation of the platform becomes. Pinduoduo, for example, has a worse reputation than JD.com. Consistent with the literature, e.g., Kastanakis and Balabanis (2011), Kung and Zhong (2017), Chiu et al. (2018), Niculescu et al. (2018), Dou and Wu (2021), we assume that a typical consumer located at x acquires a net consumption utility of $(1-\phi)v-\phi(1-\widetilde{x})\beta+\delta-t(1-x)-p_r$ from buying the product of the reseller on the platform at a price ofp_r, where the coefficient β represents the intensity of the network effects and $\phi(1-\widetilde{x})$ represents the sales of the fake product on the platform. So, we capture the negative network externality due to counterfeits on the platform via $-\phi(1-\widetilde{x})\beta$. Let the superscript E denote the results in this extension. Given p_m andp_r, we can derive the demand functions for the manufacturer's direct channel and the resell channel by solving$v-t\widetilde{x}-p_m=(1-\phi)v-\phi(1-\widetilde{x})\beta+\delta-t(1-\widetilde{x})-p_r$, as follows:

$$D_m^E=\widetilde{x}=\frac{\phi\beta+\phi v-\delta+t-p_m+p_r}{\phi\beta+2t} \tag{6}$$

$$D_r^E=1-\widetilde{x}=\frac{\delta-\phi v+t-p_r+p_m}{\phi\beta+2t} \tag{7}$$

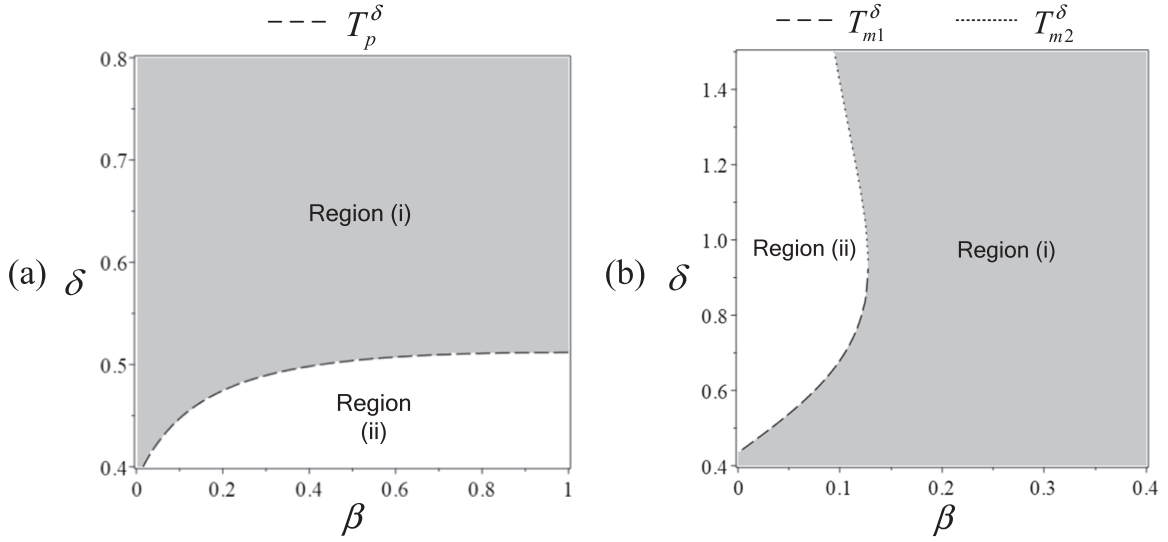


Fig. 11. The impact of anti-counterfeiting on π_p^E andπ_m^E.

The rest of the model remains the same as in the main model. Next, we characterize the equilibrium results for the model considering the network externality. The expressions of the thresholds involved in the following propositions can be found in the appendix.

Proposition 8. *The anti-counterfeiting revenues of the platform and the manufacturer:*

(i) *if* $\delta > T_p^{\delta}$, *the anti-counterfeiting revenue of the platform is positive*; *otherwise*, *it is negative*.

(ii) *if* $\delta < T_{m1}^{\delta}$ *or* $\delta > T_{m2}^{\delta}$, *the anti-counterfeiting revenue of the manufacturer is positive; otherwise, it is negative.*

Proposition 8 is illustrated in Fig. 11. We find that all the qualitative results in the main model carry over to the model considering the network externality. Specifically, as shown in Fig. 11(a) and (b), the anti-counterfeiting revenue of the platform and the manufacturer is positive in Region (i), while the anti-counterfeiting revenue of the platform and the manufacturer is negative in Region (ii).

The only notable change of result is that the manufacturer will always obtain a positive anti-counterfeiting revenue if the intensity of the network effects is high. The reason is that, if the intensity of the network effects is high, consumers have a very low valuation on the product sold by the reseller, and the reseller can only set a very low retail price, so that the manufacturer can only charge a very low wholesale price. Cracking down on the fake product has led to a sharp increase in retail price for the reseller, which in turn will benefit the manufacturer from a large increase in the wholesale price.

Proposition 9. *The incentives for the platform and manufacturer to invest in anti-counterfeiting technology*.

(i) *if* $F \leqslant \min\left\{T_m^F, T_p^F\right\}$, *both the platform and the manufacturer have incentives to invest in anti-counterfeiting technology*;

(ii) *if* $\min\left\{T_m^F, T_p^F\right\} < F \leqslant \max\left\{T_m^F, T_p^F\right\}$ *and* $\delta \leqslant T_1^{\delta}$ *or* $\delta \geqslant T_2^{\delta}$, *only the manufacturer has the incentive to invest in anti-counterfeiting technology*;

(iii) *if* $\min\left\{T_m^F, T_p^F\right\} < F \leqslant \max\left\{T_m^F, T_p^F\right\}$ *and* $T_1^{\delta} < \delta < T_2^{\delta}$, *only the platform has the incentive to invest in anti-counterfeiting technology*;

(iv) *otherwise*, *neither the platform nor the manufacturer has the incentive to invest in anti-counterfeiting technology*.

Proposition 9 is illustrated in Fig. 12. Specifically, both the platform and the manufacturer have incentives to invest in anti-counterfeiting technology in Region (i). Only the manufacturer has the incentive to invest in anti-counterfeiting technology in Region (ii), and only the platform has the incentive to invest in anti-counterfeiting technology in Region (iii). While neither the platform nor the manufacturer has the incentive to invest in anti-counterfeiting technology in Region (iv).

The only notable change of result is that the network externality increases the region in which manufacturer is willing to invest in anti-counterfeiting technology. The reason is that, compared with the main model without the network externality, the network externality increases the anti-counterfeiting revenue of the manufacturer.

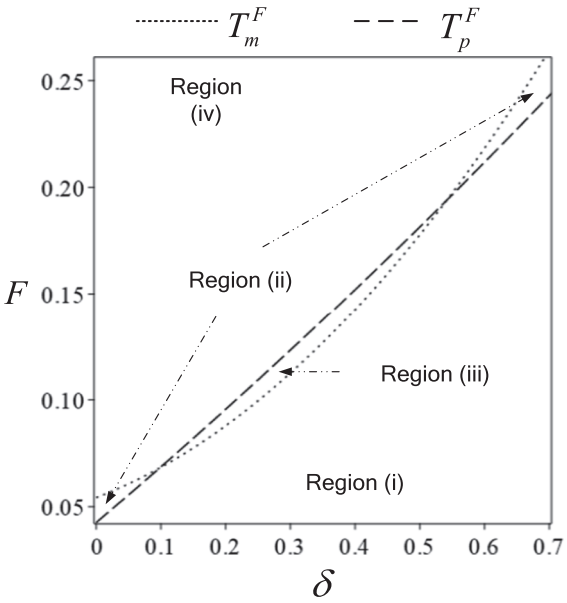


Fig. 12. The investment incentives of the manufacturer and platform in the extension.

6. Conclusions

At present, some leading retail platforms and manufacturers have adopted blockchain technology to combat counterfeiting. However, under dual channel competition, it is not clear who has a stronger incentive to invest in anti-counterfeiting technology, the platform, or the manufacturer. In this paper, we construct a dual-channel competition model composed of a manufacturer's direct channel and a reseller's resell channel on the platform, taking into account consumers' channel preferences and differences in channel services. We analyze the price competition between the manufacturer and the reseller in the cases with and without anti-counterfeiting, revealing the impact of anti-counterfeiting and the incentives for the platform and the manufacturer to invest in anti-counterfeiting technology.

In our model, the manufacturer produces and sells products directly to consumers, and she also sells products indirectly through a reseller on a retail platform. The reseller sells authentic products and fake products to consumers together through the platform. However, the retail platform and the manufacturer can choose whether to use anti-counterfeiting technology. Due to the characteristics of new anti-counterfeiting technologies such as blockchain, if the platform and the manufacturer choose to fight against counterfeiting, the reseller will no longer sell counterfeit goods.

Our study finds that even without the consideration of costs on technology investment, anti-counterfeiting is not always the dominant strategy for the manufacturer. On the one hand, cracking down on counterfeits drives down the wholesale price. On the other hand, cracking down on counterfeits enables the manufacturer to avoid the wholesale loss caused by fake product in resell channel. In addition, while cracking down on counterfeits increases the retail price of the reseller, the retail price or the demand in the direct channel may also increase. However, when the proportion of the fake product is large, the reseller is willing to pay a high wholesale price to the manufacturer due to the cost advantage of the fake product. In this scenario, anti-counterfeiting reduces the wholesale price too much, offsetting the benefits of the wholesale loss avoidance and the increase of the retail price or demand in the direct channel. We also find that the platform may suffer from anti-counterfeiting. Specifically, when the proportion of the fake product and consumers' valuation on the product are both low, anti-counterfeiting makes the demand of the reseller drop a lot, offsetting the increase of retail price of the reseller, therefore resulting in the commissions charged by the platform decreasing. Therefore, both the manufacturer and the platform cannot blindly fight against counterfeiting. In addition, we also find that the fake product does not necessarily harm consumer surplus and social welfare. So, from the government's perspective, when deciding whether to implement strict supervision on counterfeit products, the government should make a trade-off between eliminating counterfeit and avoiding damage to the revenues of the market entities.

In addition, we point out that when the investment cost of anti-counterfeiting is relatively high, either the manufacturer or the platform has an incentive to incur the investment cost, contingent on consumers' valuation on the platform's service. In line with intuition, if the platform can provide much better service that the manufacturer, only the platform has the incentive to fight against counterfeits. We find that this result is consistent with the practice. For fast moving consumer goods, it is very different for manufacturers to provide value-added services in their direct channels. Thus, the platform should play a leading role in anti-counterfeiting by adopting the new technology or helping manufacturers to do so. As the examples we mentioned in Section 1, with the help of Amazon, the world's leading retail platform, Nestlé is developing blockchain technology with which consumers are empowered to identify the authenticity of the product. However, the manufacturers of luxury goods usually are good at provide superior services. Thus, we can observe that luxury brands like LV are active in adopting new technologies to fight against counterfeits in their direct channels.

Our analysis also shows that when anti-counterfeiting is forced to be carried out under strict anti-counterfeiting regulations, if consumers' valuation on the platform's services is high, the platform can also obtain a positive revenue from anti-counterfeiting. At the same time, in the case with anti-counterfeiting, the revenue of the platform increases with the increase of consumers' valuation on the platform's services. Therefore, after using anti-counterfeiting technology to eliminate counterfeit products, the platform should provide better platform services than before. This is an important message for the platform. With the development of new technologies such as blockchain, anti-counterfeiting will be increasingly popular in the near future. Thus, even if the platform does not fight against counterfeits by itself, the manufacturer can address the problem alone. Thus, to compete with the manufacturer's direct channel, the platform with poor services like Pinduoduo must improve its attractiveness.

This is the first study examining anti-counterfeiting in a retail platform under dual channel competition, but it has several limitations that might be addressed in future research. First, in this study, we focus on the case of deceptive counterfeits; the case of non-deceptive counterfeits is equally important but less affected by anti-counterfeiting, which is worth investigating in the future research. Second, with the development of anti-counterfeiting technologies, the copycat firms may change their competitive strategies, for example, from selling deceptive counterfeits to selling non-deceptive counterfeits; Thus, it seems interesting to examine the copycat firms' reaction in the presence of anti-counterfeiting in future. Third, we consider that the platform is just a marketplace to collect commissions from the reseller. In the future, we can study the anti-counterfeiting decision of the platform when the platform also has its own private brand.

CRediT authorship contribution statement

Yu Zhou: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Xiang Gao:** Formal analysis, Writing – original draft. **Suyuan Luo:** Formal analysis, Writing – original draft, Writing – review & editing. **Yu Xiong:** Writing – review & editing. **Niangyue Ye:** Writing – review & editing.

Declaration of Competing Interest

This research was supported by the National Natural Science Foundation of China [grant number 71971033 and 72101159] and the Fundamental Research Funds for the Central Universities [grant number 2021CDJSKJC12]. Ministry of Education in China (MOE) Project of Humanities and Social Sciences, Grant/Award Number: 20YJC630092.

Appendix

We define the following thresholds.

$$t_1^v = \max\left\{\begin{array}{c}(3c+3d+11\delta+15t)\alpha^3-((11\phi+8)c+19d+143t\\ +27\delta)\alpha^2+(16c\phi+16d+16\delta+256t)\alpha-128t\\ \hline \alpha(11\alpha^2\phi-8\alpha^2-27\alpha\phi+8\alpha+16\phi)\\ (3c+3d+11\delta+15t)\alpha^3-(8c+19d+143t\\ +27\delta)\alpha^2+(16d+256t+16\delta)\alpha-128t\\ \hline 8\alpha^2(1-\alpha)\end{array}\right\},$$

$$t_2^v = c/(1-\alpha)t_1^d = \max\{v\alpha-\delta(1-\alpha)-5t,(\phi v-\delta)(1-\alpha)+v\alpha-c\phi-5t\}$$

$$t_2^d = \min\{(3t-\delta)(1-\alpha)+v\alpha,(\phi v-\delta+3t)(1-\alpha)+v\alpha-c\phi\}$$

$$t_1^\delta = \frac{8\alpha\phi v+9\alpha t-8v\alpha+8c\phi-8\phi v+8d+16t}{8(\alpha-1)}$$

$$t_1^\phi = \frac{8\alpha\delta-9\alpha t+8\alpha v-8d-8\delta-16t}{8(\alpha v+c-v)}$$

$$t_2^\phi = \frac{\begin{array}{c}4\alpha^2\delta v+4\alpha^2v^2-17\alpha v^2+8v^2+\alpha c\delta+\alpha cv-\alpha dv-18\alpha\delta v\\ -5\alpha tv+2cd-6c\delta+10tc-8cv+6vd+14v\delta+30tv\end{array}}{2(\alpha v+c-v)(2\alpha v-c-7v)}$$

$$t_2^\delta = (9\alpha t-8v\alpha+8d+16t)/(8(\alpha-1))$$

$$t_p^\delta = -\frac{4\alpha^2v^2-17v^2\alpha+8v^2+\alpha cv-\alpha dv-5\alpha tv+2cd+10ct-8cv+6dv+30tv}{4v\alpha^2+\alpha c-18v\alpha-6c+14v}$$

$$t_p^\phi = \frac{\begin{array}{c}4\alpha^2\delta v+4\alpha^2v^2-17\alpha v^2+8v^2+\alpha c\delta+\alpha cv-\alpha dv-18\alpha\delta v\\ -5\alpha tv+2cd-6c\delta+10tc-8cv+6vd+14v\delta+30tv\end{array}}{2\alpha^2v^2-9\alpha v^2+7v^2-c^2+\alpha cv-6cv}$$

$$t_m^\phi = (8\alpha\delta-9\alpha t+8\alpha v-8d-8\delta-16t)/(4(\alpha v+c-v))\quad t_m^F = \pi_m^{A^*}-\pi_m^{N^*}\quad t_p^F = \pi_p^{A^*}-\pi_p^{N^*}$$

$$t_3^\delta = \frac{\begin{array}{c}2\alpha^3\phi v^2-4v^2\alpha^3+\alpha^2c\phi v-13\alpha^2\phi v^2-\alpha^2cv+\alpha^2dv-4\alpha^2tv+25\alpha^2v^2\\ +15\alpha\phi v^2-16\alpha v^2-4\phi v^2-\alpha c^2\phi-4c^2\phi-14\alpha c\phi v-2\alpha cd-19\alpha ct\\ +16\alpha cv-14\alpha dv-37\alpha tv+8c\phi v-8cd-16ct+8dv+16vt\end{array}}{4v\alpha^3+\alpha^2c-26v\alpha^2-14\alpha c+30v\alpha+8c-8v}$$

$$t_c^\phi = (2\alpha\delta-3\alpha t+2\alpha v-2d-2\delta-2t)/(\alpha v+c-v)$$

$$t_s^\phi = \frac{\begin{array}{c}(v^2-(3c-3d-3t-4\delta)v-3c\delta)\alpha^2-(6v^2+(12d-6c\\ +16t+18\delta)v-6c(d+3t+2\delta))\alpha+2(v-c)(7d+19t+7\delta)\end{array}}{(v-\alpha v-c)(3c\alpha-2\alpha v-7c+7v)}$$

$$T_p^\delta = \frac{A_1+((3\alpha-8)t+2\phi\beta(\alpha-2))(3\alpha-8)\sqrt{A_2}}{2\beta(\alpha-1)(8\alpha^3\beta\phi+15\alpha^3t-60\alpha^2\beta\phi-121\alpha^2t+144\alpha\beta\phi+312\alpha t-112\phi\beta-256t)},\ \text{where}\ A_1$$

$$\begin{aligned}&= -2(8\beta^2\phi+t^2+(9\phi+15)t\beta)v\alpha^4+(4\phi(d+5t+33v)\beta^2-3(3c-118v)t^2+2(84v\phi+6d+21t+130v)t\beta)\alpha^3+(-4\phi(10d\\ &\quad+41t+92v)\beta^2+2(51c-623v)t^2+2(9c\phi-283v\phi-55d-173t-378v)t\beta)\alpha^2+(16\phi(7d+23t+25v)\beta^2-32t^2(11c-57v)\\ &\quad-16t(6c\phi-50v\phi-19d-51t-52v)\beta)\alpha-32(3d+7t+4v)\phi\beta^2+128t(c\phi-3v\phi-2d-4t-2v)\beta+(384c-896v)t^2\end{aligned}$$

$$,A_2=((-56\alpha^2+312\alpha-256)\beta^2+\left(4\nu\alpha^2+(c-18\nu)\alpha-6c+14\nu\right)^2-4(\alpha-1)(\nu\alpha^2+(19c+13\nu)\alpha-64c-64\nu)\beta)t^2-4(4(\alpha-1)\phi(2\alpha-7)\beta^2+2(\alpha-1)(((4\phi+1)\nu+4c\phi-d)\alpha-14\phi(c+\nu))\beta-2\nu^2\phi\alpha^4+((17\phi-1)\nu+d)\nu\alpha^3+((6-49\phi)\nu^2+2((2\phi-2)c-3d)\nu+2c(c\phi+2d))\alpha^2+((-9+55\phi)\nu^2+((-18\phi+19)c+9d)\nu-c(9c\phi+19d))\alpha+(4-21\phi)\nu^2+2((7\phi-10)c-2d)\nu+7c^2\phi+20cd)\beta t+4\beta^2(\alpha-2)^2(d-\nu)^2$$

;

$$T_{m1}^{\delta}=\frac{B_1-\sqrt{2}((3\alpha-8)t+2\phi\beta(\alpha-2))(3\alpha-8)\sqrt{B_2}}{2\beta\left((15\alpha^2-64\alpha+64)t+8\phi\beta(\alpha-2)^2\right)(\alpha-1)},T_{m2}^{\delta}=\frac{B_1+\sqrt{2}((3\alpha-8)t+2\phi\beta(\alpha-2))(3\alpha-8)\sqrt{B_2}}{2\beta\left((15\alpha^2-64\alpha+64)t+8\phi\beta(\alpha-2)^2\right)(\alpha-1)},\text{ where }B_1=-4(3\alpha-8)$$

$$\left((3\alpha^2-11\alpha+8)\nu+(3c-8\beta)\alpha-8c+8\beta\right)t^2-2((3\alpha-8)((3\phi+5)\alpha^2-(11\phi+8)\alpha-8\phi)\nu+64c\phi-64\phi\beta-64d+(9c\phi-28\phi\beta-15d)\alpha^2+(92\phi\beta-48c\phi+64d)\alpha)\beta t+16\phi\beta^2(\alpha-2)^2(d-\alpha\nu)$$

$$,B_2=t(16\phi(\alpha-1)\beta^3+\left(8\alpha^2\phi\nu+2(15t+4(c-\nu)\phi)\alpha-30t\right)\beta^2+((4\nu\phi+15t)\alpha+4(c-\nu)\phi)(\alpha\nu+c-\nu)\beta+8t(\alpha\nu+c-\nu)^2);$$

$$T_m^F=\pi_m^{AE^*}-\pi_m^{NE^*};T_p^F=\pi_p^{AE^*}-\pi_p^{NE^*};$$

$$T_1^{\delta}=\frac{C_1-(3\alpha-8)((3\alpha-8)t+2\phi\beta(\alpha-2))\sqrt{C_2}}{2\beta(\alpha-1)(8\alpha^4\beta\phi+15\alpha^4t-76\alpha^3\beta\phi-151\alpha^3t+224\alpha^2\beta\phi+470\alpha^2t-240\alpha\beta\phi-512\alpha t+64\beta\phi+128t)}T_2^{\delta}=\frac{C_1+(3\alpha-8)((3\alpha-8)t+2\phi\beta(\alpha-2))\sqrt{C_2}}{2\beta(\alpha-1)(8\alpha^4\beta\phi+15\alpha^4t-76\alpha^3\beta\phi-151\alpha^3t+224\alpha^2\beta\phi+470\alpha^2t-240\alpha\beta\phi-512\alpha t+64\beta\phi+128t)}$$

$$C_1=-2\left(8\beta^2\phi+3(3\phi+5)t\beta+18t^2\right)\nu\alpha^5+((426\nu-9c+42\beta)t^2+4((51\nu+5\beta)\phi+3d+80\nu)\beta t+4\beta^2\phi(d+41\nu))\alpha^4+((174c-1774\nu-538\beta)t^2-24(3d+22\nu)\phi\beta^2+2\beta((27c-415\nu-138)\phi-85d-536\nu)t)\alpha^3+((3176\nu-808c+1712\beta)t^2+16\phi(17d+41\nu)\beta^2-4((81c-369\nu-212\beta)\phi-155d-336\nu)\beta t)\alpha^2+((1280c-2304\nu-1728\beta)t^2-32(11d+8\nu)\phi\beta^2+16((36c-68\nu-53\beta)\phi-48d-32\nu)\beta t)\alpha+(512\nu-512c+512\beta)t^2-256((c-\nu-\beta)\phi-d)\beta t+128\beta^2d\phi$$

$$C_2=(4\alpha^2(\alpha-2)^2\beta^2+4\left(2\alpha^2-11\alpha+4\right)^2(\alpha-1)^2t^2+4(\alpha-1)^2(\left(2\alpha^2-11\alpha+4\right)(\alpha^2-5\alpha+2)\phi+\alpha^3-6\alpha^2)\beta t)\nu^2+(8(2\left(2\alpha^2-11\alpha+4\right)(1-\alpha)\alpha t\phi-\left(\alpha^3+3\alpha^2-8\alpha+4\right)t-d\alpha(\alpha-2)^2)\alpha\beta^2-4(((12\alpha^4-86\alpha^3+142\alpha^2-84\alpha+16)\phi-4\alpha^4+21\alpha^3-22\alpha^2)c+\alpha(\alpha-1)\left(\left(16\alpha^3-91\alpha^2+19\alpha+6\right)t+d\alpha(\alpha-1)(\alpha-6)\right))t\beta+4\left(2\alpha^2-11\alpha+4\right)(\alpha-1)t^2c\left(\alpha^2-14\alpha+8\right))\nu-16\left(2\alpha^2-11\alpha+4\right)(\alpha-1)\phi t(3\alpha-2)\beta^3+c^2t^2\left(\alpha^2-14\alpha+8\right)^2+(32\left(2\alpha^2-11\alpha+4\right)\alpha(1-\alpha)t\phi c+\left(-176\alpha^4+1432\alpha^3-2376\alpha^2+1360\alpha-240\right)t^2+8d\alpha(\alpha-1)\left(\alpha^2+4\alpha-4\right)t+4d^2\alpha^2(\alpha-2)^2)\beta^2-4tc(\left(2\alpha^2-11\alpha+4\right)(\alpha-1)\phi(\alpha+2)c+\alpha\left(\left(34\alpha^3-205\alpha^2+213\alpha-42\right)t+d\alpha\left(4\alpha^2-21\alpha+22\right)\right))\beta$$

Proof of Lemma 1

We first derive the reseller's optimal retail price response function. The first-order condition of π_r gives.

$$\frac{(1-\alpha)(\delta-\phi\nu+p_m-p_r+t)-(p_r(1-\alpha)-(1-\phi)w)}{2t}=0$$

We can quickly check that the second-order condition is$\frac{\partial^2\pi_r}{\partial p_r^2}=-\frac{(1-\alpha)}{t}<0$, so by solving the above equation, we can obtain that the reseller's optimal retail price response function is.

$$p_r=\frac{(1-\alpha)(\delta-\phi\nu+t+p_m)+(1-\phi)w}{2(1-\alpha)}$$

Next, we derive the manufacturer's optimal retail price response function. When choosing the retail price, the manufacturer anticipates the reseller's retail price as above. By substituting the retail price response of the reseller into the revenue of the manufacturer, we have the revenue of the manufacturer. The first-order condition of the revenue of the manufacturer with respect to p_m yields.

$$\frac{((c+\nu-w)\phi-\delta+d+3t+w-2p_m)\alpha+(2w-c-\nu)\phi+\delta-d-3t-2w+2p_m}{4t(-1+\alpha)}=0$$

We also check that the second-order condition for maximum is$\frac{\partial^2\pi_m}{\partial p_m^2}=-\frac{1}{2t}<0$, so by solving the above equation, we have the manufacturer's optimal retail price response function.

$$p_m^{N^*}(w)=\frac{(1-\alpha)((c+\nu)\phi+d+3t-\delta)+(2-\alpha)(1-\phi)w}{2(1-\alpha)}$$

Accordingly, we have.

$$p_r^{N*}(w) = \frac{(1-\alpha)((c-v)\phi + d + 5t + \delta) + (4-\alpha)(1-\phi)w}{4(1-\alpha)}$$

Proof of Proposition 1

When choosing the wholesale price, the manufacturer anticipates the retail price responses as shown in Lemma 1. By substituting the retail price responses into the revenue of the manufacturer, we have.

$$\pi_m = \left(\frac{((c+v)\phi + d + 3t - \delta)(1-\alpha) + (2-\alpha)(1-\phi)w}{2-2\alpha} - d - c\right) \times \frac{((v-c+w)\phi - d + 3t - w - \delta)\alpha + (c-v)\phi + d - 3t + \delta}{8t(\alpha-1)} + (1-\phi)\frac{((c-v-w)\phi + d + 5t + w + \delta)\alpha + (v-c)\phi - d - 5t - \delta}{8t(\alpha-1)}(w-c)$$

The second-order derivative is $\frac{\partial^2 \pi_m}{\partial w^2} = \frac{(\phi-1)^2\alpha^2}{8(\alpha-1)^2 t} > 0$, so the optimal wholesale price is either the upper bound or the lower bound of the wholesale price. Now we search for the upper bound of the wholesale price. Because the net consumption utility of consumer located at $\tilde{x}$ is non-negative, we can have $v - tD_m - p_m^{N*}(w) \geqslant 0$, which can be simplified to $w \leqslant w_u$, where $w_u = \frac{(1-\alpha)(8v-3\phi c-5\phi v-3d+5\delta-15t)}{(1-\phi)(8-3\alpha)}$. At the same time, we find that.

$$Dr = \frac{((v-c+w)\phi - d - 5t - w - \delta)\alpha + (c-v)\phi + d + 5t + \delta}{8t(1-\alpha)}$$

$$\pi_r = \frac{(((v-c+w)\phi - d - 5t - w - \delta)\alpha + (c-v)\phi + d + 5t + \delta)^2}{32t(1-\alpha)} = 2t(1-\alpha)Dr^2$$

Since we have already guaranteed $D_r > 0$ by assuming $t_1^d < d < t_2^d$ in the model, so $\pi_r > 0$ is also guaranteed. Next, in order to ensure that the manufacturer obtains non-negative marginal revenue in the resell channel, $w \geqslant c$ shall be met. So, we obtained the upper bound w_u and the lower bound c of the wholesale price. Let $\frac{\partial \pi_m}{\partial w} = 0$, we solved the stagnation point to be.

$$w_s = \frac{(\alpha-1)(\alpha c\phi - \alpha\phi v + \alpha d + \alpha\delta + 5\alpha t - 8t)}{\alpha^2(\phi-1)}$$

With the assumption $v > t_1^v$ in the model, we find that $w_u - w_s > w_s - c$. In summary, the upper bound is the optimal wholesale price. So, we obtain.

$$w^{N*} = \frac{(1-\alpha)(8v - 3\phi c - 5\phi v - 3d + 5\delta - 15t)}{(1-\phi)(8-3\alpha)}$$

Based on the optimal wholesale price, we can obtain the optimal retail prices and demands of the manufacturer and the reseller, as follows:

$$p_m^{N*} = \frac{((4-\phi)v - 3t + \delta)\alpha + (\phi-8)v - \phi c - d + 3t - \delta}{3\alpha - 8}$$

$$p_r^{N*} = \frac{((2-2\phi)v + 2\delta)\alpha + (7\phi-8)v + \phi c + d + 5t - 7\delta}{3\alpha - 8}$$

$$D_m^{N*} = \frac{(\phi v - \delta + 3t - v)\alpha + (c-v)\phi + d - 3t + \delta}{3\alpha t - 8t}$$

$$D_r^{N*} = \frac{(v - \phi v + \delta)\alpha + (v-c)\phi - d - 5t - \delta}{3\alpha t - 8t}$$

Based on the optimal wholesale price, the optimal retail prices, and demands, we can obtain the revenues of the manufacturer, the reseller, and the platform, as follows:

$$\pi_m^{N^*} = \left(\frac{((4-\phi)v-3t+\delta)\alpha+(\phi-8)v-c\phi-d+3t-\delta}{3\alpha-8}-d-c\right)\times \frac{(\phi v-\delta+3t-v)\alpha+(c-v)\phi+d-3t+\delta}{3\alpha-8t}+ (1-\phi)\frac{(v-\phi v+\delta)\alpha+(v-c)\phi-d-5t-\delta}{3\alpha-8t}\times \left(\frac{(3\phi c+5v\phi+3d-5\delta+15t-8v)(\alpha-1)}{(\phi-1)(3\alpha-8)}-c\right)$$

$$\pi_r^{N^*} = \frac{2(1-\alpha)((\phi v-\delta-v)\alpha+(c-v)\phi+d+5t+\delta)^2}{(3\alpha-8)^2 t}$$

$$\pi_p^{N^*} = \alpha\frac{((2-2\phi)v+2\delta)\alpha+(7\phi-8)v+c\phi+d+5t-7\delta}{3\alpha-8}\times \frac{(v-\phi v+\delta)\alpha+(v-c)\phi-d-5t-\delta}{(3\alpha-8)t}$$

Based on the above expressions, we can obtain the following derivatives to prove the impacts of parameters in Table 2 and Table 3, and the impacts of parameters in Table 4 can also be proved by replacing the proportion of the fake product ϕ by 0 in the following derivatives.

$\frac{\partial p_m^{N^*}}{\partial\delta} = \frac{\alpha-1}{3\alpha-8} > 0; \frac{\partial p_r^{N^*}}{\partial\delta} = \frac{2\alpha-7}{3\alpha-8} > 0; \frac{\partial D_m^{N^*}}{\partial\delta} = \frac{1-\alpha}{3\alpha t-8t} < 0; \frac{\partial D_r^{N^*}}{\partial\delta} = \frac{\alpha-1}{3\alpha t-8t} > 0; \frac{\partial p_m^{N^*}}{\partial\phi} = \frac{v-\alpha v-c}{3\alpha-8}$, so $\frac{\partial p_m^{N^*}}{\partial\phi}\geqslant 0$ if $v\leqslant\frac{c}{1-\alpha}$ and $\frac{\partial p_m^{N^*}}{\partial\phi} < 0$ if $v > \frac{c}{1-\alpha}; \frac{\partial p_r^{N^*}}{\partial\phi} = \frac{7v-2v\alpha+c}{3\alpha-8} < 0; \frac{\partial D_m^{N^*}}{\partial\phi} = \frac{\alpha v+c-v}{3\alpha t-8t}$, so $\frac{\partial p_m^{N^*}}{\partial\phi} > 0$ if $v > \frac{c}{1-\alpha}$ and $\frac{\partial p_m^{N^*}}{\partial\phi}\leqslant 0$ if $v\leqslant\frac{c}{1-\alpha}; \frac{\partial D_r^{N^*}}{\partial\phi} = \frac{v-\alpha v-c}{3\alpha-8}$, so $\frac{\partial D_r^{N^*}}{\partial\phi}\geqslant 0$ if $v\leqslant\frac{c}{1-\alpha}$ and $\frac{\partial D_r^{N^*}}{\partial\phi} < 0$ if $v > \frac{c}{1-\alpha}; \frac{\partial^2\pi_m^{N^*}}{\partial\delta^2} = \frac{8(1-\alpha)^2}{(3\alpha-8)^2 t} > 0$, and let $\frac{\partial\pi_m^{N^*}}{\partial\delta} = 0$, we can obtain $\delta = t_1^\delta$, so $\frac{\partial\pi_m^{N^*}}{\partial\delta} > 0$ if $\delta > t_1^\delta$ and $\frac{\partial\pi_m^{N^*}}{\partial\delta}\leqslant 0$ if $\delta\leqslant t_1^\delta; \frac{\partial\pi_r^{N^*}}{\partial\delta} = \frac{4(\alpha-1)^2 D_r^{N^*}}{8-3\alpha} > 0$; for $\frac{\partial D_r^{N^*}}{\partial\delta} > 0$ and $\frac{\partial p_r^{N^*}}{\partial\delta} > 0$, we can obtain $\frac{\partial\pi_p^{N^*}}{\partial\delta} > 0$; $\frac{\partial^2\pi_m^{N^*}}{\partial\phi^2} = \frac{8(v\alpha+c-v)^2}{(3\alpha-8)^2 t} > 0$, and let $\frac{\partial\pi_m^{N^*}}{\partial\phi} = 0$, we can obtain $\phi = t_1^\phi$, so $\frac{\partial\pi_m^{N^*}}{\partial\phi} > 0$ if $\phi > t_1^\phi$ and $\frac{\partial\pi_m^{N^*}}{\partial\phi}\leqslant 0$ if $\phi\leqslant t_1^\phi; \frac{\partial\pi_r^{N^*}}{\partial\phi} = \frac{4(\alpha-1)(\alpha v+c-v)D_r^{N^*}}{3\alpha-8}$, so $\frac{\partial\pi_r^{N^*}}{\partial\phi} > 0$ if $v < \frac{c}{1-\alpha}$ and $\frac{\partial\pi_r^{N^*}}{\partial\phi}\leqslant 0$ if $v\geqslant\frac{c}{1-\alpha}$. If $v < \frac{c}{1-\alpha}, \frac{\partial^2\pi_p^{N^*}}{\partial\phi^2} = \frac{\alpha(7v-2\alpha v+c)(v-\alpha v-c)}{(3\alpha-8)^2 t} < 0$, and let $\frac{\partial\pi_p^{N^*}}{\partial\phi} = 0$ we can obtain $\phi = t_2^\phi$, so $\frac{\partial\pi_p^{N^*}}{\partial\phi} > 0$ *if* $\phi < t_2^\phi$ and $\frac{\partial\pi_p^{N^*}}{\partial\phi}\leqslant 0$ *if* $\phi\geqslant t_2^\phi$; if $v\geqslant\frac{c}{1-\alpha}$, for $\frac{\partial D_r^{N^*}}{\partial\phi}\leqslant 0$ and $\frac{\partial p_r^{N^*}}{\partial\phi}\leqslant 0$, we can obtain $\frac{\partial\pi_p^{N^*}}{\partial\phi}\leqslant 0$.

Proof of Proposition 2

First, we prove the impact of anti-counterfeiting on the retail price of the reseller. In the case without anti-counterfeiting, the retail price of the reseller is.

$$p_r^{N^*} = \frac{((2-2\phi)v+2\delta)\alpha+(7\phi-8)v+\phi c+d+5t-7\delta}{3\alpha-8}$$

In the case with anti-counterfeiting, the retail price of the reseller is.

$$p_r^{A^*} = \frac{(2\delta+2v)\alpha+d-7\delta+5t-8v}{3\alpha-8}$$

So, we can have $p_r^{A^*}-p_r^{N^*} = ((7v-2v\alpha+c)\phi)/(8-3\alpha) > 0$.

Next, we prove the impact of anti-counterfeiting on the retail price of the manufacturer. In the case without anti-counterfeiting, the retail price of the manufacturer is.

$$p_m^{N^*} = \frac{((4-\phi)v-3t+\delta)\alpha+(\phi-8)v-\phi c-d+3t-\delta}{3\alpha-8}$$

In the case with anti-counterfeiting, the retail price of the manufacturer is.

$$p_m^{A^*} = \frac{(4v-3t+\delta)\alpha-8v-d+3t-\delta}{3\alpha-8}$$

So, we can have $p_m^{A^*}-p_m^{N^*} = ((v-\alpha v-c)\phi)/(8-3\alpha)$. Now, we can quickly obtain the following cases:

(i) if $v > \frac{c}{1-\alpha}, p_m^{A^*} > p_m^{N^*}$;
(ii) if $v\leqslant\frac{c}{1-\alpha}, p_m^{A^*}\leqslant p_m^{N^*}$.

Finally, we prove the impact of anti-counterfeiting on the demands. In the case without anti-counterfeiting, the demands of the manufacturer and the reseller are.

$$D_m^{N^*} = \frac{(\phi v - \delta + 3t - v)\alpha + (c - v)\phi + d - 3t + \delta}{3\alpha t - 8t}$$

$$D_r^{N^*} = \frac{(v - \phi v + \delta)\alpha + (v - c)\phi - d - 5t - \delta}{3\alpha t - 8t}$$

In the case with anti-counterfeiting, the demands of the manufacturer and the reseller are.

$$D_m^{A^*} = \frac{(3t - \delta - v)\alpha + d - 3t + \delta}{3\alpha t - 8t}$$

$$D_r^{A^*} = \frac{(\delta + v)\alpha - d - 5t - \delta}{3\alpha t - 8t}$$

So, we can have $D_r^{A^*} - D_r^{N^*} = ((v - v\alpha - c)\phi)/(8t - 3\alpha t)$ and $D_m^{A^*} - D_m^{N^*} = ((\alpha v + c - v)\phi)/(8t - 3\alpha t)$. Now, we can quickly obtain the following cases:

(i) if $v \leqslant \frac{c}{1-\alpha}$, $D_r^{A^*} \leqslant D_r^{N^*}$ and $D_m^{A^*} \geqslant D_m^{N^*}$;

(ii) if $v > \frac{c}{1-\alpha}$, $D_r^{A^*} > D_r^{N^*}$ and $D_m^{A^*} < D_m^{N^*}$.

Proof of Proposition 3

In the case without anti-counterfeiting, the revenue of the platform is.

$$\pi_p^{N^*} = \alpha \frac{((2 - 2\phi)v + 2\delta)\alpha + (7\phi - 8)v + c\phi + d + 5t - 7\delta}{3\alpha - 8} \times \frac{(v - \phi v + \delta)\alpha + (v - c)\phi - d - 5t - \delta}{(3\alpha - 8)t}$$

In the case with anti-counterfeiting, the revenue of the platform is.

$$\pi_p^{A^*} = \alpha \frac{(2v + 2\delta)\alpha - 8v + d + 5t - 7\delta}{3\alpha - 8} \frac{(\delta + v)\alpha - d - 5t - \delta}{(3\alpha - 8)t}$$

So, we can have.

$$\pi_p^{A^*} - \pi_p^{N^*} = \frac{\begin{array}{c}\alpha(((\alpha - 1)v + c)((7 - 2\alpha)v + c))\phi^2 + \alpha((4\alpha^2 \\ -17\alpha + 8)v^2 + (4\alpha^2\delta + (c - d - 5t - 18\delta)\alpha - 8c \\ +6d + 30t + 14\delta)v + c(\alpha\delta + 2d - 6\delta + 10t))\phi\end{array}}{(3\alpha - 8)^2 t}$$

Let $\pi_p^{A^*} - \pi_p^{N^*} = 0$, we can obtain $\phi = 0$ and $\phi = t_p^{\phi}$. When $v \leqslant \frac{c}{1-\alpha}$, $\frac{\partial^2(\pi_p^{A^*} - \pi_p^{N^*})}{\partial\phi^2} = \frac{\alpha(((\alpha-1)v+c)((7-2\alpha)v+c))}{(3\alpha-8)^2 t} \geqslant 0$, so $\pi_p^{A^*} > \pi_p^{N^*}$ ($\pi_p^{A^*} \leqslant \pi_p^{N^*}$) if $\phi > t_p^{\phi}$ ($\phi \leqslant t_p^{\phi}$). When $v > \frac{c}{1-\alpha}$, we can obtain $D_r^{A^*} > D_r^{N^*}$ and $p_r^{A^*} - p_r^{N^*} > 0$, so $\pi_p^{A^*} > \pi_p^{N^*}$.

Now, we can obtain the following cases:

(i) if $v > \frac{c}{1-\alpha}$ or $\phi > t_p^{\phi}$, $\pi_p^{A^*} > \pi_p^{N^*}$;

(ii) if $v \leqslant \frac{c}{1-\alpha}$ and $\phi \leqslant t_p^{\phi}$, $\pi_p^{A^*} \leqslant \pi_p^{N^*}$;

Proof of Proposition 4

In the case without anti-counterfeiting, the revenue of the manufacturer is.

$$\pi_m^{N^*} = \left(\frac{((4 - \phi)v - 3t + \delta)\alpha + (\phi - 8)v - c\phi - d + 3t - \delta}{3\alpha - 8} - d - c\right) \times \frac{(\phi v - \delta + 3t - v)\alpha + (c - v)\phi + d - 3t + \delta}{3\alpha - 8t} + (1 - \phi)\frac{(v - \phi v + \delta)\alpha + (v - c)\phi - d - 5t - \delta}{3\alpha - 8t} \times \left(\frac{(3\phi c + 5v\phi + 3d - 5\delta + 15t - 8v)(\alpha - 1)}{(\phi - 1)(3\alpha - 8)} - c\right)$$

In the case with anti-counterfeiting, the revenue of the platform is.

$$\pi_p^{A^*} = \left(\frac{(4v-3t+\delta)\alpha-8v-d+3t-\delta}{3\alpha-8}-d-c\right)\frac{(3t-\delta-v)\alpha+d-3t+\delta}{3\alpha-8t} + \frac{(\delta+v)\alpha-d-5t-\delta}{3\alpha-8t}\left(\frac{(3d-5\delta+15t-8v)(\alpha-1)}{8-3\alpha}-c\right)$$

So, we can have.

$$\pi_m^{A^*}-\pi_m^{N^*} = -\frac{4(\alpha v+c-v)^2\phi^2+((-8v+9t-8\delta)\alpha+8d+16t+8\delta)(\alpha v+c-v)\phi}{(3\alpha-8)^2 t}$$

Obviously, $\pi_m^{A^*}-\pi_m^{N^*}$ is concave with respect toϕ. Let$\pi_m^{A^*}-\pi_m^{N^*}=0$, we can obtain $\phi=0$ and$\phi=t_m^{\phi}$.
Now, we can obtain the following cases:

(i) if$\phi \geqslant t_m^{\phi}$,$\pi_m^{A^*} \leqslant \pi_m^{N^*}$;
(ii) if$\phi < t_m^{\phi}$,$\pi_m^{A^*} > \pi_m^{N^*}$.

Proof of Proposition 5

In this part, we will prove that $t_m^F < t_p^F$ ($t_m^F \geqslant t_p^F$) if $\delta > t_3^{\delta}$ ($\delta \leqslant t_3^{\delta}$). Note that t_p^F and t_m^F represent the anti-counterfeiting revenue of the platform and of the manufacturer, respectively. By simplifying, we can obtain.

$$t_p^F - t_m^F = \frac{\begin{array}{c}(4v\alpha^3+(c-26v)\alpha^2+(-14c+30v)\alpha+8c-8v)\phi\delta+(-2v^2(\phi-2)\alpha^3\\ +((13\phi-25)v^2+((-\phi+1)c-d+4t)v)\alpha^2+((-15\phi+16)v^2+((14\phi\\ -16)c+14d+37t)v+c(c\phi+2d+19t))\alpha+4(c-v)(c\phi-\phi v+2d+4t))\phi\end{array}}{(3\alpha-8)^2 t}$$

To focus on the more interesting result, we assume that$v < ((-\alpha^2+14\alpha-8)c)/(2(2\alpha^3-13\alpha^2+15\alpha-4))$, which means that v will not be too large. So, we can prove that$4v\alpha^3+(c-26v)\alpha^2+(-14c+30v)\alpha+8c-8v>0$, which means that $t_p^F - t_m^F$ increases withδ. Let$t_p^F - t_m^F = 0$, we can have$\delta = t_3^{\delta}$. So, we proved that $t_p^F - t_m^F > 0$ if $\delta > t_3^{\delta}$ and $t_p^F - t_m^F \leqslant 0$ if$\delta \leqslant t_3^{\delta}$.
Now, we can obtain the following cases:

(i) if$F \leqslant \min\left\{t_m^F, t_p^F\right\}$, both the platform and the manufacturer gain more from anti-counterfeiting than the cost of the anti-counterfeiting technology;
(ii) if $\min\left\{t_m^F, t_p^F\right\} < F \leqslant \max\left\{t_m^F, t_p^F\right\}$ and$\delta \leqslant t_3^{\delta}$, only the manufacturer gains more from anti-counterfeiting than the cost of the anti-counterfeiting technology;
(iii) if $\min\left\{t_m^F, t_p^F\right\} < F \leqslant \max\left\{t_m^F, t_p^F\right\}$ and$\delta > t_3^{\delta}$, only the platform gains more from anti-counterfeiting than the cost of the anti-counterfeiting technology;
(iv) if$F > \max\left\{t_m^F, t_p^F\right\}$, both platform and manufacturer gain less from anti-counterfeiting than the cost of the anti-counterfeiting technology.

Proof of Proposition 6

In the case without anti-counterfeiting, the consumer surplus is.

$$CS^{N*} = \frac{\begin{array}{c}(2\phi^2v^2+2(3t-2v-2\delta)v\phi+2v^2+(4\delta-6t)v+9t^2-6\delta t+2\delta^2)\alpha^2\\ +(4v(c-v)\phi^2+(4v^2+(4d-4c-2t+8\delta)v+2c(3t-2\delta))\phi\\ -4(d+t+\delta)v-18t^2+(6d+2\delta)t-4\delta(d+\delta))\alpha+2(c-v)^2\phi^2\\ +4(d+t+\delta)(c-v)\phi+34t^2+(4d+4\delta)t+2(d+\delta)^2\end{array}}{2t(3\alpha-8)^2}$$

In the case with anti-counterfeiting, the consumer surplus is.

$$CS^{A*} = \frac{\begin{array}{c}(2v^2+(4\delta-6t)v+9t^2-6\delta t+2\delta^2)\alpha^2-(4(d+t+\delta)v+18t^2\\ -(6d+2\delta)t+4\delta(d+\delta))\alpha+34t^2+(4d+4\delta)t+2(d+\delta)^2\end{array}}{2t(3\alpha-8)^2}$$

So, we can have.

$$CS^{N*}-CS^{A*} = \frac{(\alpha v+c-v)^2\phi^2+(\alpha v+c-v)((-2v+3t-2\delta)\alpha+2d+2t+2\delta)\phi}{t(3\alpha-8)^2}$$

Obviously, $CS^{N*} - CS^{A*}$ is convex with respect to ϕ. Let $CS^{N*} - CS^{A*} = 0$, we can obtain $\phi = 0$ and $\phi = t_c^{\phi}$.
Now, we can quickly obtain the following cases:

(i) if $\phi \geqslant t_c^{\phi}$, $CS^{A*} \leqslant CS^{N*}$;
(ii) if $\phi < t_c^{\phi}$, $CS^{A*} > CS^{N*}$.

Proof of Proposition 7

In the case without anti-counterfeiting, the social welfare is $SW^{N*} = \pi_p^{N*} + \pi_m^{N*} + \pi_r^{N*} + CS^{N*}$. In the case with anti-counterfeiting, the social welfare is $SW^{A*} = \pi_p^{A*} + \pi_m^{A*} + \pi_r^{A*} + CS^{A*}$. By simplifying, we can obtain.

$$SW^{N*} - SW^{A*} = -\frac{\begin{array}{c}(\alpha v + c - v)(3c\alpha - 2\alpha v - 7c + 7v)\phi^2 + ((v^2 + (3d - 3c \\ +3t + 4\delta)v - 3c\delta)\alpha^2 - (6v^2 - (6c - 12d - 16t - 18\delta)v \\ -c(6d + 12\delta + 18t))\alpha - (c - v)(14d + 38t + 14\delta))\phi\end{array}}{(3\alpha - 8)^2 t}$$

Let $SW^{N*} - SW^{A*} = 0$, we can obtain $\phi = 0$ and $\phi = t_s^{\phi}$. Note that $v > c$ is always true. Let $3c\alpha - 2\alpha v - 7c + 7v = 0$, we obtain $v = ((3\alpha - 7)c)/(2\alpha - 7)$. Because $((3\alpha - 7)c)/(2\alpha - 7) - c = c\alpha/(2\alpha - 7) \langle 0$, so $3c\alpha - 2\alpha v - 7c + 7v > 0$ is true with $v > c$. So, when $v \leqslant \frac{c}{1-\alpha}$, $\frac{\partial^2 (SW^{N*} - SW^{A*})}{\partial \phi^2} = \frac{-(\alpha v + c - v)(3c\alpha - 2\alpha v - 7c + 7v)}{(3\alpha - 8)^2 t} \leqslant 0$, therefore $SW^{A*} > SW^{N*}$ ($SW^{A*} \leqslant SW^{N*}$) if $\phi > t_s^{\phi}$ ($\phi \leqslant t_s^{\phi}$). When $v > \frac{c}{1-\alpha}$, $\frac{\partial^2 (SW^{N*} - SW^{A*})}{\partial \phi^2} = \frac{-(\alpha v + c - v)(3c\alpha - 2\alpha v - 7c + 7v)}{(3\alpha - 8)^2 t} > 0$. We will prove that $t_s^{\phi} > 1$ when $v > \frac{c}{1-\alpha}$ in the following.

We denote the numerator and the denominator of t_s^{ϕ} as M_n and M_d respectively. Let $\psi(v) = M_n - M_d$, we can obtain $\psi(v) = \begin{array}{c}(-\alpha^2 + 3\alpha - 7)v^2 - 3\alpha^2 c\delta + 3c(c + 2d + 6t + 4\delta)\alpha - c(7c + 14d + 38t + 14\delta) \\ +((3d + 3t + 4\delta)\alpha^2 + (-6c - 12d - 16t - 18\delta)\alpha + 14d + 38t + 14\delta + 14c)v\end{array}$ and $\frac{\partial^2 \psi(v)}{\partial v^2} = -\alpha^2 + 3\alpha - 7 < 0$. First, we will prove that $\psi(v) \geqslant 0$ when v is the lower bound. When $v > \frac{c}{1-\alpha}$, the lower bound of v is $v = \frac{c}{1-\alpha}$ and $\psi(v = c/(1-\alpha)) = ((-\alpha^2\delta + (c + d + 2\delta + 5t)\alpha - d - 5t - \delta)(3\alpha - 8)c\alpha)/(\alpha - 1)^2$, and we can find that $\psi(v = c/(1-\alpha))$ increases with d. For $D_r^{A*} > 0$, we can obtain $d > \alpha\delta + \alpha v - \delta - 5t$. When $d = \alpha\delta + \alpha v - \delta - 5t$, we find that $\psi(v = c/(1-\alpha)) = 0$. So, $\psi(v = c/(1-\alpha)) \geqslant 0$ is true. Next, we will prove that $\psi(v) \geqslant 0$ when v is the upper bound. In order to ensure that $D_r^{N*} > 0$ for any ϕ, we can obtain $v < -\alpha\delta + c + d + \delta + 5t$. So, the upper bound of v is $v = -\alpha\delta + c + d + \delta + 5t$. For $\psi(v = c/(1-\alpha)) \geqslant 0$, We can obtain $-\alpha^2\delta + (c + d + 2\delta + 5t)\alpha - d - 5t - \delta \leqslant 0$ and so $\begin{array}{c}\psi(v = -\alpha\delta + c + d + \delta + 5t) = -(-\alpha^2\delta + (c + d + 2\delta + 5t)\alpha \\ -d - 5t - \delta)(-\alpha^2\delta + (c - 2d + 2t - \delta)\alpha + 7d + 3t + 7\delta) \geqslant 0\end{array}$.

In summary, when $v > \frac{c}{1-\alpha}$, we proved that $\psi(v) \geqslant 0$ when v is the lower bound or the upper bound and $\frac{\partial^2 \psi(v)}{\partial v^2} < 0$. So, when $v > \frac{c}{1-\alpha}$, $\psi(v) = M_n - M_d \geqslant 0$ is true, therefore we proved $t_s^{\phi} > 1$.
Now, we can obtain the following cases:

(i) if $v > \frac{c}{1-\alpha}$ or $\phi > t_s^{\phi}$, $SW^{A*} > SW^{N*}$;
(ii) if $v \leqslant \frac{c}{1-\alpha}$ and $\phi \leqslant t_s^{\phi}$, $SW^{A*} \leqslant SW^{N*}$;

Proof of Proposition 8

First, we derive the optimal wholesale and retail prices in the case without anti-counterfeiting. When considering the network externality, the calculation process is consistent with the main model, so we omit it. The optimal wholesale and retail prices and demands of the manufacturer and the reseller, as follows:

$$w^{NE*} = \frac{\begin{array}{c}(1 - \alpha)(4\phi^2\beta^2 + 2\beta c\phi^2 + 2\beta\phi^2 v + 2\beta d\phi - 2\beta\delta\phi \\ +16\beta\phi t - 4\beta\phi v + 3c\phi t + 5\phi t v + 3dt - 5\delta t + 15t^2 - 8tv)\end{array}}{(1 - \phi)(2\phi\beta\alpha + 3\alpha t - 4\phi\beta - 8t)}$$

$$p_m^{NE*} = \frac{\begin{array}{c}(3 - 3\alpha)t^2 + ((v - (v + 2\beta)\alpha - c + 2\beta)\phi \\ +(4v + \delta)\alpha - d - 8v - \delta)t + 2v\phi\beta(\alpha - 2)\end{array}}{t(3\alpha - 8) + 2\beta(\alpha - 2)\phi}$$

$$p_r^{NE*} = \frac{\begin{array}{c}\beta(2\beta - \alpha v + c + 3v)\phi^2 + ((7t + (v + \delta)\alpha + d - 4v - 3\delta)\beta \\ +t(c - 2\alpha v + 7v))\phi + t(5t + (2v + 2\delta)\alpha + d - 8v - 7\delta)\end{array}}{t(3\alpha - 8) + 2\beta(\alpha - 2)\phi}$$

$$D_m^{NE*} = \frac{((v + 2\beta)\phi + 3t - v - \delta)\alpha + (c - v - 2\beta)\phi + d - 3t + \delta}{(2\phi\beta + 3t)\alpha - 4\phi\beta - 8t}$$

$$D_r^{NE^*} = \frac{(v-\alpha v-2\beta-c)\phi+(v+\delta)\alpha-d-5t-\delta}{2\beta(\alpha-2)\phi+3\alpha t-8t}$$

Based on the optimal wholesale price, the optimal retail prices, and demands, we can obtain the revenues of the manufacturer, the reseller, and the platform, as follows:

$$\pi_m^{NE^*} = \left(\begin{array}{c} (3-3\alpha)t^2+((v-(v+2\beta)\alpha-c+2\beta)\phi \\ \frac{+(4v+\delta)\alpha-d-8v-\delta)t+2v\phi\beta(\alpha-2)}{t(3\alpha-8)+2\beta(\alpha-2)\phi}-d-c \end{array} \right) \times$$
$$\frac{((v+2\beta)\phi+3t-v-\delta)\alpha+(c-v-2\beta)\phi+d-3t+\delta}{(2\phi\beta+3t)\alpha-4\phi\beta-8t}+$$
$$(1-\phi)\frac{(v-\alpha v-2\beta-c)\phi+(v+\delta)\alpha-d-5t-\delta}{2\beta(\alpha-2)\phi+3\alpha t-8t}\times$$
$$\left(\begin{array}{c} (1-\alpha)(4\phi^2\beta^2+2\beta c\phi^2+2\beta\phi^2 v+2\beta d\phi-2\beta\delta\phi \\ \frac{+16\beta\phi t-4\beta\phi v+3c\phi t+5\phi tv+3dt-5\delta t+15t^2-8tv)}{(1-\phi)(2\phi\beta\alpha+3\alpha t-4\phi\beta-8t)}-c \end{array} \right)$$

$$\pi_r^{NE^*} = \left(\begin{array}{c} \beta(2\beta-\alpha v+c+3v)\phi^2+((7t+(v+\delta)\alpha+d-4v-3\delta)\beta \\ (1-\alpha)\frac{+t(c-2\alpha v+7v))\phi+t(5t+(2v+2\delta)\alpha+d-8v-7\delta)}{t(3\alpha-8)+2\beta(\alpha-2)\phi} \\ (1-\alpha)(4\phi^2\beta^2+2\beta c\phi^2+2\beta\phi^2 v+2\beta d\phi-2\beta\delta\phi \\ -\frac{+16\beta\phi t-4\beta\phi v+3c\phi t+5\phi tv+3dt-5\delta t+15t^2-8tv)}{(2\phi\beta\alpha+3\alpha t-4\phi\beta-8t)} \end{array} \right)$$
$$\times\frac{(v-\alpha v-2\beta-c)\phi+(v+\delta)\alpha-d-5t-\delta}{2\beta(\alpha-2)\phi+3\alpha t-8t}$$

$$\pi_p^{NE^*} = \alpha\frac{\beta(2\beta-\alpha v+c+3v)\phi^2+((7t+(v+\delta)\alpha+d-4v-3\delta)\beta}{+t(c-2\alpha v+7v))\phi+t(5t+(2v+2\delta)\alpha+d-8v-7\delta)}{t(3\alpha-8)+2\beta(\alpha-2)\phi}$$
$$\times\frac{(v-\alpha v-2\beta-c)\phi+(v+\delta)\alpha-d-5t-\delta}{2\beta(\alpha-2)\phi+3\alpha t-8t}$$

In the case with anti-counterfeiting, the specific expressions can be obtained by replacing the proportion of the fake product ϕ by 0 in the equilibrium results of the case without anti-counterfeiting.

We denote.

$$T_{po}^{\delta} = \frac{A_1-((3\alpha-8)t+2\phi\beta(\alpha-2))(3\alpha-8)\sqrt{A_2}}{2\beta(\alpha-1)(8\alpha^3\beta\phi+15\alpha^3 t-60\alpha^2\beta\phi-121\alpha^2 t} \\ +144\alpha\beta\phi+312\alpha t-112\phi\beta-256t)$$

Let$\pi_p^{NE^*}-\pi_p^{AE^*}=0$, we can obtain $\delta=T_p^{\delta}$ and$\delta=T_{po}^{\delta}$. We find that.

$$\frac{\partial^2\left(\pi_p^{NE^*}-\pi_p^{AE^*}\right)}{\partial\delta^2} = \frac{\alpha(1-\alpha)\beta\phi(8\alpha^3\beta\phi+15\alpha^3 t-60\alpha^2\beta\phi}{-121\alpha^2 t+144\alpha\beta\phi+312\alpha t-112\beta\phi-256t)}{(2\alpha\beta\phi+3\alpha t-4\beta\phi-8t)^2(3\alpha-8)^2 t} < 0$$

At the same time, for$D_r^{AE^*}>0$, we can obtain$\delta>(\alpha v-d-5t)/(1-\alpha)$. By comparing, we have$T_{po}^{\delta}<(\alpha v-d-5t)/(1-\alpha)$. So, $\delta=T_p^{\delta}$ is the unique solution. Therefore, we proved that $\pi_p^{NE^*}<\pi_p^{AE^*}$ if$\delta>T_p^{\delta}$, and $\pi_p^{NE^*}\geqslant\pi_p^{AE^*}$ if $\delta\leqslant T_p^{\delta}$ otherwise.

Let$\pi_m^{NE^*}-\pi_m^{AE^*}=0$, we can obtain $\delta=T_{m1}^{\delta}$ and$\delta=T_{m2}^{\delta}$. We find that$-32\alpha\beta\phi-64\alpha t+32\beta\phi+64t>0$, so$8(\alpha-4)\alpha\beta\phi+15\alpha^2 t-64\alpha t+32\beta\phi+64t>0$, and therefore we obtain.

$$\frac{\partial^2\left(\pi_m^{NE^*}-\pi_m^{AE^*}\right)}{\partial\delta^2} = -\frac{2\beta\phi(\alpha-1)^2(8(\alpha-4)\alpha\beta\phi+15\alpha^2 t-64\alpha t+32\beta\phi+64t)}{(2\alpha\beta\phi+3\alpha t-4\beta\phi-8t)^2(3\alpha-8)^2 t} < 0$$

So, we proved that $\pi_m^{NE^*}<\pi_m^{AE^*}$ if $\delta<T_{m1}^{\delta}$ or$\delta>T_{m2}^{\delta}$, $\pi_m^{NE^*}\geqslant\pi_m^{AE^*}$ if $T_{m1}^{\delta}\leqslant\delta\leqslant T_{m2}^{\delta}$ otherwise.

Proof of Proposition 9

In this part, we will prove that $T_m^F < T_p^F$ if$T_1^\delta < \delta < T_2^\delta$, $T_m^F \geqslant T_p^F$ if $\delta \leqslant T_1^\delta$ or$\delta \geqslant T_2^\delta$. Note that T_p^F and T_m^F represent the anti-counterfeiting revenue of the platform and of the manufacturer in the model considering the network externality, respectively. Let$T_m^F - T_p^F = 0$, we can obtain $\delta = T_1^\delta$ and$\delta = T_2^\delta$. With the assumption $\alpha < 0.2$ made in the model, we find that.

$$\frac{\partial^2\left(T_m^F - T_p^F\right)}{\partial\delta^2} = \frac{\begin{array}{c}\beta\phi(-8\alpha^5\beta\phi - 15\alpha^5 t + 84\alpha^4\beta\phi + 166\alpha^4 t - 300\alpha^3\beta\phi - 621\alpha^3 t \\ +464\alpha^2\beta\phi + 982\alpha^2 t - 304\alpha\beta\phi - 640\alpha t + 64\beta\phi + 128t)\end{array}}{(2\alpha\beta\phi + 3\alpha t - 4\beta\phi - 8t)^2 t(3\alpha - 8)^2} > 0$$

So, we proved that $T_m^F < T_p^F$ if$T_1^\delta < \delta < T_2^\delta$, $T_m^F \geqslant T_p^F$ if $\delta \leqslant T_1^\delta$ or$\delta \geqslant T_2^\delta$.

Now, we can obtain the following cases:

(i) if$F \leqslant \min\left\{T_m^F, T_p^F\right\}$, both platform and manufacturer gain more from anti-counterfeiting than the cost of the anti-counterfeiting technology;

(ii) if $\min\left\{T_m^F, T_p^F\right\} < F \leqslant \max\left\{T_m^F, T_p^F\right\}$ and $\delta \leqslant T_1^\delta$ or$\delta \geqslant T_2^\delta$, only the manufacturer gains more from anti-counterfeiting than the cost of the anti-counterfeiting technology;

(iii) if $\min\left\{T_m^F, T_p^F\right\} < F \leqslant \max\left\{T_m^F, T_p^F\right\}$ and$T_1^\delta < \delta < T_2^\delta$, only the platform gains more from anti-counterfeiting than the cost of the anti-counterfeiting technology;

(iv) if$F > \max\left\{T_m^F, T_p^F\right\}$, both platform and manufacturer gain less from anti-counterfeiting than the cost of the anti-counterfeiting technology.

References

Abhishek, V., Jerath, K., Zhang, Z.J., 2016. Agency selling or reselling? Channel structures in electronic retailing. Manage. Sci. 62 (8), 2259–2280.

Arya, A., Mittendorf, B., 2018. Bricks-and-mortar entry by online retailers in the presence of consumer sales taxes. Manage. Sci. 64 (11), 5220–5233.

Babich, V., Hilary, G., 2020. OM Forum—distributed ledgers and operations: what operations management researchers should know about blockchain technology. Manuf. Service Operations Manage. 22 (2), 223–240.

Bae, J., Chen, L., Yao, S., 2022. Service capacity and price promotion wars. Manage. Sci., forthcoming.. https://doi.org/10.1287/mnsc.2022.4325.

Cao, K., Xu, X., Bian, Y., Sun, Y., 2019. Optimal trade-in strategy of business-to-consumer platform with dual-format retailing model. Omega 82, 181–192.

Cao, Y., Shen, B., 2022. Adopting blockchain technology to block less sustainable products' entry in global trade. Transp. Res. Part E: Logistics Transp. Rev. 161, 102695.

Chiu, C.-H., Choi, T.-M., Dai, X., Shen, B., Zheng, J.-H., 2018. Optimal advertising budget allocation in luxury fashion markets with social influences: a mean-variance analysis. Production Operations Manage. 27 (8), 1611–1629.

Cho, S.-H., Fang, X., Tayur, S., 2015. Combating Strategic Counterfeiters in Licit and Illicit Supply Chains. Manuf. Service Operations Manage. 17 (3), 273–289.

Choi, T.-M., 2019. Blockchain-technology-supported platforms for diamond authentication and certification in luxury supply chains. Transp. Res. Part E: Logistics Transp. Rev. 128, 17–29.

Choi, T.-M., Chen, Y., Chung, S.H., 2019a. Online-offline fashion franchising supply chains without channel conflicts: Choices on postponement and contracts. Int. J. Prod. Econ. 215, 174–184.

Choi, T.-M., Feng, L., Li, R., 2020a. Information disclosure structure in supply chains with rental service platforms in the blockchain technology era. Int. J. Prod. Econ. 221, 107473.

Choi, T.-M., Guo, S., Liu, N., Shi, X., 2019b. Values of food leftover sharing platforms in the sharing economy. Int. J. Prod. Econ. 213, 23–31.

Choi, T.-M., Guo, S., Luo, S., 2020b. When blockchain meets social-media: Will the result benefit social media analytics for supply chain operations management? Transp. Res. Part E: Logistics Transp. Rev. 135, 101860.

Choi, T.-M., He, Y., 2019. Peer-to-peer collaborative consumption for fashion products in the sharing economy: Platform operations. Transp. Res. Part E: Logistics Transp. Rev. 126, 49–65.

Choi, T.-M., Luo, S., 2019. Data quality challenges for sustainable fashion supply chain operations in emerging markets: Roles of blockchain, government sponsors and environment taxes. Transp. Res. Part E: Logistics Transp. Rev. 131, 139–152.

Choi, T.-M., Ouyang, X., 2021. Initial coin offerings for blockchain based product provenance authentication platforms. Int. J. Prod. Econ. 233, 107995.

Choi, T.-M., Wen, X., Sun, X., Chung, S.-H., 2019c. The mean-variance approach for global supply chain risk analysis with air logistics in the blockchain technology era. Transp. Res. Part E: Logistics Transp. Rev. 127, 178–191.

Cui, Q., 2019. Quality investment, and the contract manufacturer's encroachment. Eur. J. Oper. Res. 279 (2), 407–418.

Cui, Y., Duenyas, I., Şahin, Ö., 2014. Should event organizers prevent resale of tickets? Manage. Sci. 60 (9), 2160–2179.

Dong, C., Yang, Y., Zhao, M., 2018. Dynamic selling strategy for a firm under asymmetric information: direct selling vs. agent selling. Int. J. Prod. Econ. 204, 204–213.

Dou, Y., Wu, D.J., 2021. Platform Competition Under Network Effects: Piggybacking and Optimal Subsidization. Information Syst. Res. 32 (3), 820–835.

Gao, S.Y., Lim, W.S., Tang, C.S., 2017a. Entry of copycats of luxury brands. Marketing Sci. 36 (2), 272–289.

Gao, S.Y., Lim, W.S., Tang, C.S., 2017b. The impact of the potential entry of copycats: entry conditions, consumer welfare, and social welfare. Decision Sci. 48 (4), 594–624.

Gao, Y., 2018. On the use of overt anti-counterfeiting technologies. Marketing Sci. 37 (3), 403–424.

Ghamat, S., Pun, H., Critchley, G., Hou, P., 2021. Using intellectual property agreements in the presence of supplier and third-party copycatting. Eur. J. Oper. Res. 291 (2), 680–692.

Hagiu, A., Wright, J., 2015. Marketplace or Reseller? Manage. Sci. 61 (1), 184–203.

He, B., Gupta, V., Mirchandani, P., 2021. Online selling through O2O platform or on your own? Strategic implications for local Brick-and-Mortar stores. Omega 103, 102424.

Hu, M., Milner, J., Wu, J., 2016. Liking and following and the newsvendor: operations and marketing policies under social influence. Manage. Sci. 62 (3), 867–879.

Jerath, K., Netessine, S., Veeraraghavan, S.K., 2010. Revenue management with strategic customers: last-minute selling and opaque selling. Manage. Sci. 56 (3), 430–448.
Karp, L., Perloff, J.M., 2005. When promoters like scalpers. J. Econ. Manage. Strategy 14 (2), 477–508.
Kastanakis, M., Balabanis, G., 2011. Bandwagon, snob and veblen effects in luxury consumption. Adv. Consum. Res. 38, 609–611.
Kung, L.-C., Zhong, G.-Y., 2017. The optimal pricing strategy for two-sided platform delivery in the sharing economy. Transp. Res. Part E: Logistics Transp. Rev. 101, 1–12.
Kwark, Y., Chen, J., Raghunathan, S., 2017. Platform or wholesale? A strategic tool for online retailers to benefit from third-party information. MIS Quart. 41 (3), 763–785.
Li, G., Li, L., Sethi, S.P., Guan, X., 2019. Return strategy and pricing in a dual-channel supply chain. Int. J. Prod. Econ. 215, 153–164.
Li, G., Xue, J., Li, N., Ivanov, D., 2022. Blockchain-supported business model design, supply chain resilience, and firm performance. Transp. Res. Part E: Logistics Transp. Rev. 163, 102773.
Li, K.J., 2021. Behavior-Based Quality Discrimination. Manuf. Service Operations Manage. 23 (2), 267–545.
Liu, C., Dan, Y., Dan, B., Xu, G., 2020a. Cooperative strategy for a dual-channel supply chain with the influence of free-riding customers. Electron. Commer. Res. Appl. 43, 101001.
Liu, H., Lei, M., Deng, H., Keong Leong, G., Huang, T., 2016. A dual channel, quality-based price competition model for the WEEE recycling market with government subsidy. Omega 59, 290–302.
Liu, H., Sun, S., Lei, M., Deng, H., Leong, G.K., 2017. The impact of retailers' alliance on manufacturer's profit in a dual-channel structure. Int. J. Prod. Res. 55 (22), 6592–6607.
Liu, W., Yan, X., Li, X., Wei, W., 2020b. The impacts of market size and data-driven marketing on the sales mode selection in an Internet platform based supply chain. Transp. Res. Part E: Logistics Transp. Rev. 136, 101914.
Liu, Y., Li, J., Ren, W., Forrest, J.-Y.-L., 2020c. Differentiated products pricing with consumer network acceptance in a dual-channel supply chain. Electron. Commer. Res. Appl. 39, 100915.
Luo, H., Zhong, L., Nie, J., 2022. Quality and distribution channel selection on a hybrid platform. Transp. Res. Part E: Logistics Transp. Rev. 163, 102750.
Niculescu, M.F., Wu, D.J., Xu, L., 2018. Strategic intellectual property sharing: competition on an open technology platform under network effects. Information Syst. Res. 29 (2), 498–519.
Nie, J., Wang, Q., Shi, C., Zhou, Y., 2022. The dark side of bilateral encroachment within a supply chain. J. Operational Res. Soc. 73 (4), 811–821.
Niu, B., Mu, Z., Cao, B., Gao, J., 2021. Should multinational firms implement blockchain to provide quality verification? Transp. Res. Part E: Logistics Transp. Rev. 145, 102121.
Niu, B., Xu, H., Chen, L., 2022. Creating all-win by blockchain in a remanufacturing supply chain with consumer risk-aversion and quality untrust. Transp. Res. Part E: Logistics Transp. Rev. 163, 102778.
Pun, H., DeYong, G.D., 2017. Competing with copycats when customers are strategic. Manuf. Service Operations Manage. 19 (3), 403–418.
Pun, H., Swaminathan, J.M., Hou, P., 2021. Blockchain adoption for combating deceptive counterfeits. Production Operations Manage. 30 (4), 864–882.
Qian, Y., 2014. Brand management and strategies against counterfeits. J. Econ. Manage. Strategy 23 (2), 317–343.
Qian, Y., Gong, Q., Chen, Y., 2015. Untangling searchable and experiential quality responses to counterfeits. Marketing Sci. 34 (4), 522–538.
Qin, J., Ren, L., Xia, L., Wang, Z., Chang, H., 2020a. Pricing strategies for dual-channel supply chains under a trade credit policy. Int. Trans. Operational Res. 27 (5), 2469–2508.
Qin, X., Liu, Z., Tian, L., 2020b. The strategic analysis of logistics service sharing in an e-commerce platform. Omega 92, 102153.
Shen, B., Dong, C., Minner, S., 2022. Combating copycats in the supply chain with permissioned blockchain technology. Production Operations Manage. 31 (1), 138–154.
Shen, B., Xu, X., Yuan, Q., 2020. Selling secondhand products through an online platform with blockchain. Transp. Res. Part E: Logistics Transp. Rev. 142, 102066.
Siqin, T., Choi, T.-M., Chung, S.-H., 2022. Optimal E-tailing channel structure and service contracting in the platform era. Transp. Res. Part E: Logistics Transp. Rev. 160, 102614.
Song, W., Chen, J., Li, W., 2021. Spillover effect of consumer awareness on third parties' selling strategies and retailers' platform openness. Information Syst. Res. 32 (1), 172–193.
Song, W., Li, W., Geng, S., 2020. Effect of online product reviews on third parties' selling on retail platforms. Electron. Commer. Res. Appl. 39, 100900.
Su, X., 2010. Optimal Pricing with Speculators and Strategic Consumers. Manage. Sci. 56 (1), 25–40.
Sun, J., Zhang, X., Zhu, Q., 2020. Counterfeiters in Online Marketplaces: Stealing Your Sales or Sharing Your Costs. J. Retail. 96 (2), 189–202.
Sun, X., Tang, W., Zhang, J., Chen, J., 2021. The impact of quantity-based cost decline on supplier encroachment. Transp. Res. Part E: Logistics Transp. Rev. 147, 102245.
Tian, L., Vakharia, A.J., Tan, Y.R., Xu, Y., 2018. Marketplace, reseller, or hybrid: strategic analysis of an emerging E-commerce model. Production Operations Manage. 27 (8), 1595–1610.
Wang, X., Wang, X., Yu, B., Zhang, S., 2019. A comparative study of entry mode options for E-commerce platforms and suppliers. Electron. Commer. Res. Appl. 37, 100888.
Wei, J., Lu, J., Zhao, J., 2020. Interactions of competing manufacturers' leader-follower relationship and sales format on online platforms. Eur. J. Oper. Res. 280 (2), 508–522.
Xiao, T., Choi, T.-M., 2009. Purchasing choices and channel structure strategies for a two-echelon system with risk-averse players. Int. J. Prod. Econ. 120 (1), 54–65.
Xiao, T., Shi, J., 2016. Pricing and supply priority in a dual-channel supply chain. Eur. J. Oper. Res. 254 (3), 813–823.
Xu, X., Choi, T.-M., 2021. Supply chain operations with online platforms under the cap-and-trade regulation: Impacts of using blockchain technology. Transp. Res. Part E: Logistics Transp. Rev. 155, 102491.
Yan, N., Liu, Y., Xu, X., He, X., 2020. Strategic dual-channel pricing games with e-retailer finance. Eur. J. Oper. Res. 283 (1), 138–151.
Yang, L., Tang, R., 2019. Comparisons of sales modes for a fresh product supply chain with freshness-keeping effort. Transp. Res. Part E: Logistics Transp. Rev. 125, 425–448.
Yang, W., Zhang, J., Yan, H., 2021. Impacts of online consumer reviews on a dual-channel supply chain. Omega 101, 102266.
Yao, S., Zhu, K., 2020. Combating product label misconduct: The role of traceability and market inspection. Eur. J. Oper. Res. 282 (2), 559–568.
Yi, Z., Yu, M., Cheung, K.L., 2020. Impacts of Counterfeiting on a Global Supply Chain. Manuf. Service Operations Manage. 24 (1), 159–178.
Zennyo, Y., 2020. Strategic contracting and hybrid use of agency and wholesale contracts in e-commerce platforms. Eur. J. Oper. Res. 281 (1), 231–239.
Zhang, J., Cao, Q., He, X., 2019. Contract and product quality in platform selling. Eur. J. Oper. Res. 272 (3), 928–944.
Zhang, J., Zhang, R.Q., 2015. Supply chain structure in a market with deceptive counterfeits. Eur. J. Oper. Res. 240 (1), 84–97.
Zhang, S., Zhang, J., 2020. Agency selling or reselling: E-tailer information sharing with supplier offline entry. Eur. J. Oper. Res. 280 (1), 134–151.
Zhao, X., Guo, H., Cai, G., Bandyopadhyay, S., 2021. The role of expectation-reality discrepancy in service contracts. Production Operations Manage. 30 (11), 4160–4175.
Zhou, J., Zhao, R., Wang, B., 2020. Behavior-based price discrimination in a dual-channel supply chain with retailer's information disclosure. Electron. Commer. Res. Appl. 39, 100916.